

**[B16 ENG 1101]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**ENGLISH**  
MODEL QUESTION PAPER  
(Common to all branches)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. a) Write an Essay on **One** of the following. (7M)
  - i. Pros and cons of Internet
  - ii. Terrorism, a social evil
- b) Correct any **Five** of the following sentences. (5M)
  - i. The machineries were expensive.
  - ii. Suppose, if you arrive late, you will miss the show.
  - iii. Choose the best of the two options.
  - iv. I enjoyed during the holidays.
  - v. I have seen him yesterday.
  - vi. The teacher gave us many advices.
- c) Use the appropriate articles in the given blanks. (2M)
  - i. He speaks ..... English very well.
  - ii. I saw.....movie last night.
  - iii. Did you get married after leaving .....university?
  - iv. I was at.....train station when you called me.
2. a) Write a report on **One** of the following. (7M)
  - i. Write a feasibility report for setting up a Water / Power Unit at your campus.
  - ii. Write a report on Educational Tour
- b) Write one word substitutions to any **Four** of the following and write sentences by using them. (5M)
  - i. Language which is confusing and unintelligible.
  - ii. One who prepares plans for buildings.
  - iii. A great lover of books
  - iv. A person in charge of a museum
  - v. A man who thinks only for himself
  - vi. One who kills animals and sells their flesh
- c) Write appropriate quantifiers for each sentence (Some, few, much, lesser, a little, less). (2M)
  - i. There were ..... at the college last year
  - ii. The project is ..... complicated than the last one

- iii. I have to buy .....pairs of blue and black jeans soon.  
iv. How .....cash do you need to purchase this CD player
3. a) Write a letter on **One** of the following. (8M)
- Write a letter to a renowned person, requesting him to be the Chief Guest for the cultural festival of your college.
  - Write a letter to the editor about the problem of brain drain.
- b) Identify the types of the following sentences and write a similar sentence for each type. (4M)
- Oh, what a beautiful morning!
  - Eat your supper.
  - Today is my birthday.
  - What gifts did you receive for your birthday?
- c) Re-write the sentences by using Gerunds, to-infinitives or plain infinitive forms. (2M)
- She is good at..... (dance).
  - He is crazy about..... (sing).
  - He'd like..... (fly) an aeroplane.
  - I enjoy..... (write) picture postcards.
4. a) Draft an E-Mail to your friend about your career plans. (8M)
- b) Punctuate the following sentences taken from the text correctly. (4M)
- Sunil Sharma is Documentation Development Manager at Cerner Corporation one of the world's largest medical software developers
  - As part of his job Sunil writes web-based content for Cerner
  - One type of website that Cerner develops is marketed to health facilities for use by doctors nurses hospital administrators and patients
  - This explains the communication challenge that Sunil faces. Cerner's end user is diverse consisting of lay readers and high-tech specialists
- c) Pick the right synonyms of the following words. (2M)
- Euphoria  
a) Sober b) High spirits c) Mean d) Feeble
  - Vicious  
a) cruel b) kind c) splendid d) dearest
  - Remnant  
a) horror b) whole sale c) left over d) energize
  - Acclaim  
a) praise b) blame c) honour d) criticism
5. a) Develop a paragraph (150 words) based on the following hints. (7M)
- As the 11th President of India---- the Indian National Congress-----  
'people's president', he was----- His contribution -----Bharat Ratna. During -  
-----in India. He is the -----India: 2020 and Ignited Minds.
- b) Fill in the blanks with the appropriate idioms from the box. (5M)
- (The cream of the crop, an arm and a leg, hand in glove, hue and cry, Eager beaver, shape up)
- Frank always tries to finish his work before everyone else. He is an\_\_\_\_\_.
  - We chose the prettiest, best behaved puppy. She was certainly \_\_\_\_\_.
  - If Madge doesn't\_\_\_\_\_, she could lose her job.

- iv. Our new office was very expensive. It cost\_\_\_\_\_.
- v. The two friends are \_\_\_\_\_ with each other.
- c) Pick the right antonyms of the following words. (2M)
- Awake  
a) alive b) stir c) asleep d) truce
  - Create  
a) build b) beak c) deny d) refuse
  - Emerge  
a) abandon b) appear c) fall d) hide
  - Warm  
a) cold b) pleasant c) unkind d) indifferent
6. a) Draft a pamphlet on any Electronic home appliances/Places of tourists' interest/an Educational institution/ an exhibition. (8M)
- b) Fill in the blanks using the appropriate forms of verbs given in the brackets. (4M)
- The wind \_\_\_\_\_ furiously. (Blow)
  - He \_\_\_\_\_ to his mother every week. (Write)
  - In a fit of rage, she \_\_\_\_\_ up the letter. (Tear)
  - We couldn't have \_\_\_\_\_ a better day for organizing the party. (Choose)
- c) Fill in the blanks with appropriate prepositions from the box (in , at, the, at, on,). (2M)
- They are staying at \_\_\_\_\_ hotel
  - That is \_\_\_\_\_ girl I told you about
  - My birthday is \_\_\_\_\_ May
  - We are going to see my parents \_\_\_\_\_ the weekend
7. a) Present an argument in about 150 words on 'Women are not suitable to work in the industry.' Substantiate your argument with reasons. (7M)
- b) Read the following paragraph and answer the questions: (5M)

The study of history provides many benefits. First, we learn from the past. We may repeat mistakes, but, at least, we have the opportunity to avoid them. Second, history teaches us what questions to ask about the present. Contrary to some people's view, the study of history is not the memorization of names, dates, and places. It is the thoughtful examination of the forces that have shaped the courses of human life. We can examine events from the past and then draw inferences about current events. History teaches us about likely outcomes.

Another benefit of the study of history is the broad range of human experience which is covered. War and peace are certainly covered as are national and international affairs. However, matters of culture (art, literature, and music) are also included in historical study. Human nature is an important part of history: emotions like passion, greed, and insecurity have influenced the shaping of world affairs. Anyone who thinks that the study of history is boring has not really studied history.

- What is the main idea of this passage?
- In the first paragraph, 'inferences' mean?
- Which method of teaching history would the author of this passage support?
- In the second paragraph, 'shaping of world affairs' Means.
- What is the conclusive thought of the author?

- c) Fill the blanks by using appropriate conjunctions (because, neither-nor, and, and) (2M)
- i. Receptionists must be able to relay information \_\_\_\_\_ pass messages accurately.
  - ii. Mary is a member of the Historical Society \_\_\_\_\_ the Literary Society.
  - iii. Susie \_\_\_\_\_ phoned \_\_\_\_\_ wrote after she left home.
  - iv. The committee rejected the proposal \_\_\_\_\_ they did not think it was practical.

8. a) Select appropriate words from the below word list to complete the following sentences. (6M)  
(popularity, interact, networking, revolutionized, overwhelmed, reputation)

- i. Sachin's \_\_\_\_\_ was evidence of the fact that he was a friendly and fun to be with.
- ii. \_\_\_\_\_ is the key to understanding the market better.
- iii. Leela was \_\_\_\_\_ with emotion at the award ceremony.
- iv. His failure to reach the meeting on time has not done any good to his \_\_\_\_\_.
- v. A tiny little box between the electric guitar and the amplifier \_\_\_\_\_ rock music.
- vi. Javed said 'We at DSIJ love to \_\_\_\_\_ with our readers and we have some special sections for all of you.

- b) Write a conversation between two/ three friends who are discussing an idea for a business they would like to set up. (8M)

(or)

Write a conversation between two students discussing a social issue.

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[B16 ENG 1101]

**[B16 ENG 1102]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**MATHEMATICS - I**  
MODEL QUESTION PAPER  
(Common to all branches)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Solve the following. [ 7x2 = 14 marks]
- (a) Find the total derivative of  $x^2y$  with respect to  $x$  when  $x$  and  $y$  are connected by the relation  $x^2 + xy + y^2 = 1$
- (b) If the plane  $3x + 12y - 6z - 17 = 0$  touches the conicoid  $3x^2 - 6y^2 + 9z^2 + 17 = 0$  find the point of contact
- (c) Write the necessary conditions for  $f(x,y)$  to have a maximum or minimum at  $(a,b)$ .
- (d) Form the differential equation from the equation  $x = a \sin(\omega t + b)$
- (e) Solve  $(y^2 e^{xy^2} + 4x^3)dx + (2xy e^{xy^2} - 3y^2)dy = 0$
- (f) Solve  $\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$
- (g) Express  $f(x) = \frac{x}{2}$  as a Fourier series in the interval  $-\pi < x < \pi$

2. (a) If  $U = \tan^{-1} \frac{x^3 + y^3}{x - y}$  and  $x U_x + y U_y = \sin 2U$ , prove that

$$x^2 U_{xx} + 2xy U_{xy} + y^2 U_{yy} = 2 \cos 3U \sin U$$

(b) If  $u = x^2 - 2y^2$ ;  $v = 2x^2 - y^2$  where,  $x = r \cos \theta$ ,  $y = r \sin \theta$

show that  $\frac{\partial(u,v)}{\partial(r,\theta)} = 6 r^3 \sin 2\theta$

3. (a) Expand  $x^2 y + 3y - 2$  in powers of  $(x - 1)$  and  $(y + 2)$  using Taylor's theorem.  
 (b) By using the method of differentiation under the integral sign

Prove that  $\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)} dx = \frac{\pi}{2} \log(1+a) : a \geq 0$

4. (a) Solve  $\frac{dy}{dx} = y \tan x - y^2 \sec x$

(b) Solve  $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$

5. (a) Find the orthogonal trajectories of the family of parabolas  $ay^2 = x^3$

- (b) If 30 % of radio active substance disappeared in 10 days, how long will it take for 90 % of the substance to disappear?

6. (a) Solve  $\frac{d^2 y}{dx^2} + 4y + 5y = -2 \cosh x$  given that  $y = 0$  and  $\frac{dy}{dx} = 1$  at  $x = 0$

(b) Solve  $\frac{d^2 y}{dx^2} + 4y = \tan 2x$ , by using method of variation of parameters.

7. (a) Solve  $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$

(b) Solve the simultaneous equations  $\frac{dx}{dt} + y = \sin t$ ,  $\frac{dy}{dt} + x = \cos t$ , given that  $x = 2$  and

$y = 0$  when  $t = 0$

8. (a) Find the Fourier series of  $f(x) = x - x^2$  in the interval  $-\pi < x < \pi$

(b) Find the half-range cosine series for  $f(x) = x$  in the interval  $0 < x < 2$

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[B16 ENG 1102]

**[B16 ENG 1103]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**MATHEMATICS - II**  
MODEL QUESTION PAPER  
(Common to all branches)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Solve the following. [ 7x2 = 14 marks]
  - a) Find the value of  $\lambda$  for which the system of equations  $2x + y + 2z = 0$ ,  
 $x + y + 3z = 0$ ,  $4x + 3y + \lambda z = 0$  have a non-zero solution.
  - b) Define Hermitian matrix and give an example.
  - c) Write any two properties of Laplace transforms
  - d) Find the Laplace transform of unit step function
  - e) Find  $L^{-1}\left(\frac{s^2 - 3s + 4}{s^3}\right)$ .
  - f) Solve the difference equation  $u_{n+1} - 2u_n + 2u_{n-1} = 0$ .
  - g) Find the z-transform of  $n^2$ .
2.
  - a) Find the rank of the matrix  $A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$  by reducing into normal form.
  - b) Find the eigen values and eigen vectors of the matrix  $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ .
3.
  - a) Verify Cayley-Hamilton theorem for the matrix  $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$  and use it to evaluate the matrix equation  $A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$ .

- b) If  $A = \begin{bmatrix} 0 & 1+2i \\ -1+2i & 0 \end{bmatrix}$  then show that  $(I-A)(I+A)^{-1}$  is a unitary matrix.
4. a) Reduce the quadratic form  $2xy + 2xz - 2yz$  to canonical form by an orthogonal transformation and discuss its nature.  
 b) Solve:  $x + 2y + 3z = 14$ ,  $2x + 3y + 4z = 20$ ,  $3x + 4y + z = 14$  by Gauss elimination method.
5. a) Find i)  $L\left\{\frac{\cos at - \cos bt}{t}\right\}$  ii)  $L\left\{\int_0^t e^{-t} \cos t dt\right\}$ .  
 b) Find the Laplace transform of the triangular wave function of period  $2a$  given by  
 $f(t) = t, 0 < t < a$   
 $= 2a - t, a < t < 2a.$
6. a) Evaluate: i)  $L^{-1}\left\{\log\left(\frac{s+1}{s-1}\right)\right\}$  ii)  $L^{-1}\left\{\frac{3s}{s^2 + 2s - 8}\right\}$ .  
 b) State Convolution theorem and use it to evaluate  $L^{-1}\left\{\frac{1}{(s-2)(s+2)^2}\right\}$ .
7. a) Solve the difference equation  $y_{n+2} - 6y_{n+1} + 8y_n = 2^n$ .  
 b) Use z-transforms to solve  $y_{n+2} - 5y_{n+1} + 6y_n = 1$ , given  $y_0 = 0, y_1 = 1$ .
8. a) Find inverse Z-transform of  $\frac{z^2 + 2z}{(z+1)(z-1)^2}$  by the use of Partial fractions.  
 b) Given  $Z(u_n) = \frac{2z^2 + 3z + 4}{(z-3)^3}; |z| > 3$ , find the values of  $u_1, u_2$  and  $u_3$ .

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[B16 ENG 1103]



**[B16 ENG 1104]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**CHEMISTRY**  
MODEL QUESTION PAPER  
(Common to CIVIL, CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - (a) What is hardness of water.
  - (b) How solids are classified?
  - (c) What are ceramics?
  - (d) What is the Galvanic corrosion?
  - (e) What do you mean by conducting polymers?
  - (f) Define cetane number?
  - (g) Write any two advantages of LPG as motor fuel.
2.
  - (a) Describe the ion exchange process of water softening
  - (b) Describe the steps involved in municipal water treatment.
3.
  - (a) Write the manufacture of Portland cement.
  - (b) Write the properties and applications of ceramics.
4.
  - (a) Give a detailed account on band theory of solids.
  - (b) Explain zone refining of solids with neat diagram.
5.
  - (a) What is corrosion ? Explain the theory of dry corrosion with examples.
  - (b) What is paint? Explain the constituents of paint.
6.
  - (a) Define polymerization. Explain the mechanism of addition polymersation with suitable Examples.
  - (b) Write the preparation and properties of cellulose derivatives.
7.
  - (a) Describe the manufacture of coke by Otto- Hoffmann's process
  - (b) What is synthetic petrol? Explain Fischer Tropsch, method with a neat diagram.
8.
  - (a) Explain the desalination of water by reverse osmosis method.
  - (b) Explain the principles of Lubrication with neat diagram.

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**[B16 ENG 1104]**

**[B16 ENG 1105]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**PHYSICS**  
MODEL QUESTION PAPER  
(Common ECE, EEE & Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - (a) Distinguish between heat and work.
  - (b) What is a cyclic process and how it can be represented ?
  - (c) What is Hall effect ?
  - (d) Explain the principle of super position.
  - (e) Explain the principle of light propagation in an optical fiber.
  - (f) Define magneto striction effect.
  - (g) State the uncertainty principle.
2.
  - (a) Distinguish between reversible and irreversible process. Mention the conditions of reversibility of a process (4M)
  - (b) State and prove the Carnot's theorem (7M)
  - (c) What is the efficiency of a Carnot engine operating between melting point and boiling Point of water under normal conditions. (3M)
3.
  - (a) State and Explain the Biot and Savart law. Using it, deduce an expression for the magnetic Induction along the axis of a circular current carrying coil. (10M)
  - (b) What are Maxwell's equations and explain their significance. (4M)
4.
  - (a) Define interference phenomena of light. (2M)
  - (b) Deduce the conditions for maxima and minima of monochromatic light reflected from a thin transparent film. (8M)
  - (c) Describe the characteristics of lasers. (4M)
5.
  - (a) Define numerical aperture of an optical fiber and what is its physical significance. (2M)
  - (b) Deduce an expression for the numerical aperture of a fiber (7M)
  - (c) Mention the important applications of ultra sonics (5M)

6. (a) What are matter waves and describe their properties . (3M)  
(b) Deduce the Schrodingers time independent wave equation. (8M)  
(c) Give a classification of materials based on the band theory of solids (3M)
7. (a) What are nano materials and describe the methods of characterizing the nano materials (6M)  
(b) Describe with neat figure, any one method of synthesis of nano materials. (8M)
8. Write about  
(a) Entropy and disorder (4M)  
(b) Requirements of any laser device (4M)  
(c) Piezoelectric method of producing ultrasonics (6M)

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**[B16 ENG 1105]**

**[B16 ENG 1106]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**COMPUTER PROGRAMMING USING C & NUMERICAL METHODS**  
MODEL QUESTION PAPER  
(Common to CIVIL, CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

- 1 Write a short answer to the following. . [ 7x2 = 14 marks]
- a) What is recursion? Give an example.
  - b) Explain scope and extent of variables.
  - c) What are truncation and round off errors?
  - d) Distinguish between local and global variables.
  - e) Explain different bitwise operators?
  - f) Explain Euler's Method.
  - g) What is Interpolation?
- 2 a) Explain different types of operators in C.
- b) Write a program to check whether the given number is palindrome or not.
- 3 a) What is an array? Explain two dimensional array with an example?
- b) Write a C program to generate prime numbers less than the given number.
- 4 a) What is a Pointer? How is it initialized? What is the function of a pointer variable? What are its uses?
- b) What is a loop ?Explain general forms of all loop structures with suitable examples.
- 5 a) Explain the difference between call by reference & call by value?
- b)Write a program to sort an array of elements in ascending order?

6 a) Explain the following

i) Structure

ii) Accessing elements in structure

iii) Arrays of structures

b) Briefly explain file handling functions.

7 a) Use gauss elimination method to solve

$$2x+y+z=10, 3x+2y+3z=18, x+4y+9z=16$$

b) Given  $y' = y - x$ , where  $y(0)=2$  find  $y(0.1)$  and  $y(0.2)$  using Runge-kutta fourth order method

8 a) Find the root of the following equation using Newton-Raphson method, correct the result upto 3 decimal places.

$$X^3 - 3X - 5 = 0.$$

b) Evaluate

$$\int_{-2}^2 x \sin(x) dx \text{ using Simpson's } 1/3 \text{ rule.}$$

$$-2$$

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**[B16 ENG 1106]**

**[B16 ENG 1107]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**ENGINEERING GRAPHICS**  
**MODEL QUESTION PAPER**  
(Common to ECE, EEE & Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. . [ 7x2 = 14 marks]
  - (a) What is an involute? Write its uses?
  - (b) Define Conics.
  - (c) What is an auxiliary plane? State its purpose?
  - (d) Define frustum of a solid.
  - (e) Define the term section plane.
  - (f) State methods of developments.
  - (g) Define isometric scale.
2. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight.
3. A line AB, of 80 mm long has its end A, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP.
4. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground.
5. A pentagonal pyramid, with base 30mm and height 80mm, rests on one edge of its base on HP. The highest point in the base is 30mm above HP. Draw its projections, when the axis is parallel to VP. Drawn another front view, on a reference line inclined at  $30^0$  to the edge on which it is resting, so that the base is visible.
6. A cone, base 75 mm diameter and axis 80 mm long is resting on its base on the H.P. it is cut by a section plane perpendicular to the V.P., inclined at  $45^0$  to the H.P. and cutting the axis at a point 35 mm from the apex. Draw its front view, sectional top view and true shape of the section.

7. A right regular hexagonal pyramid of 30 mm side of base and height of 70 mm stands with its base on HP. A through circular hole of 30 mm diameter is drilled through the pyramid such that the axis of the hole is perpendicular to VP and intersects the axis of the pyramid 20 mm above the base. Draw the development of the lateral surface of the pyramid showing the true shape of the holes formed on it.
8. A right circular cylinder 5 cm diameter of base and 7 cm height has its base in the HP. A right circular cone diameter of base 4 cm and height 4 cm rests centrally over the upper flat surface of the cylinder. Draw the isometric view of the above combination

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**[B16 ENG 1107]**

**[B16 ENG 1108]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**HISTORY OF SCIENCE AND TECHNOLOGY**  
MODEL QUESTION PAPER  
(Common to CIVIL, CSE & IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) Explain the terms Science and Technology.
  - b) Describe the role of Scientist in the society.
  - c) Science and Technology Policy resolutions.
  - d) Defense Spin-offs.
  - e) Biosensors.
  - f) Barriers of Technological change.
  - g) Types of Technology transfer.
2. Describe the roots of science and technology in ancient period in India.
3. Explain the salient features of new technology fund and programs aimed at technological self reliance.
4. Describe the achievements of Council of Scientific and Industrial Research.
5. Explain the salient features of Space program and INSAT services.
6. Explain the importance of Nuclear energy and describe the effects of nuclear explosion and India's safety measures.
7. Describe the importance of Ocean development and explain the marine research and capacity building.
8. What is Appropriate technology? Explain the criteria for selection of an appropriate technology.

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**[B16 ENG 1108]**



**[B16 ENG 1109]**  
I/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**PROFESSIONAL ETHICS AND MORAL VALUES**  
MODEL QUESTION PAPER  
(Common to ECE, EEE & Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - (a) Ethical Vision
  - (b) Profession and Professionalism
  - (c) Environmental Ethics
  - (d) Bhopal Gas Tragedy
  - (e) Gender discrimination
  - (f) Cyber Crimes
  - (g) Engineers as Managers
2. Discuss the scope and aim of Engineering Ethics.
3. Explain the role of Engineers in promoting ethical climate.
4. What are Values? Explain in detail the classification of human values.
5. Elucidate the moral responsibility of engineers towards safety and risk.
6. Define the concept of globalization and explain the role of MNCs in our country.
7. What are the functions of various sample codes of ethics?
8. Discuss the need to focus on professional ethics.

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**[B16 ENG 1109]**

**[B16 ENG 1201]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**MATHEMATICS-III**  
MODEL QUESTION PAPER  
(Common to all branches)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Solve the following. [ 7x2 = 14 marks]
  - (a) Find the angle between the line  $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$  and the plane  $3x + y + z = 7$ .
  - (b) Define right circular cylinder.
  - (c) Change the integral  $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$  into polar coordinates.
  - (d) Express  $\int_0^{\frac{\pi}{2}} \sqrt{\tan \theta} d\theta$  in terms of gamma function.
  - (e) Evaluate  $\int_0^1 \int_0^{1-y} xy dx dy$  using Dirichlet's integral.
  - (f) State Parseval's identity for Fourier transforms.
  - (g) Find the Fourier cosine transform of  $f(x) = e^{-ax}$  ( $a > 0$ ).
2. (a) Find the image of the point (2, -1, 3) in the plane  $3x - 2y - z - 9 = 0$ .  
(b) Find the equation of the plane which passes through the point (3, -3, 1) and is perpendicular to the planes  $7x + y + 2z = 6$  and  $3x + 5y - 6z = 8$ .
3. (a) Prove that the three planes  $2x + y + z = 3$ ,  $x - y + 2z = 4$ ,  $x + z = 2$  form a triangular prism  
(b) Find the magnitude and equations of the shortest distance between the lines
$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+3}{4} \text{ and } \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$
4. (a) Find the equation of the sphere having its centre on the plane  $4x - 5y - z = 3$

and passing through the circle  $x^2 + y^2 + z^2 - 2x - 3y + 4z + 8 = 0$ ,  $x - 2y + z = 8$ .

(b) Find the equation of the right circular cone generated by rotating the line

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} \text{ about the line } \frac{x}{-1} = \frac{y}{1} = \frac{z}{2}.$$

5. (a) Evaluate the integral by changing the order of integration  $\int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy$ .

(b) Find by double integration the area of the lemniscate  $r^2 = a^2 \cos 2\theta$ .

6. (a) Evaluate the integral  $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z dz dx dy$ .

(b) Find the centroid of the area enclosed by the parabola  $y^2 = 4ax$ , the x-axis and its latus rectum.

7. (a) Express the function  $f(x) = \begin{cases} 1 & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases}$

as a Fourier integral. Hence evaluate  $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$ .

(b) Find the Fourier Sine transform of  $\frac{e^{-ax}}{x}$ .

8. (a) Find Fourier transform of  $f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$ .

Hence evaluate  $\int_0^\infty \frac{\sin ax}{x} dx$ .

(a) Use Parseval's identity to show that  $\int_0^\infty \frac{dt}{(t^2+1)(t^2+4)} = \frac{\pi}{12}$ .

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[B16 ENG 1201]

**[B16 ENG 1202]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**PHYSICS**  
MODEL QUESTION PAPER  
(Common to CIVIL, CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) Distinguish between heat and work.
  - b) What is a cyclic process and how it can be represented ?
  - c) What is Hall effect ?
  - d) Explain the principle of super position.
  - e) Explain the principles of light propagation in an optical fiber.
  - f) Define magneto striction effect.
  - g) State the uncertainty principle.
2.
  - (a) Distinguish between reversible and irreversible process. Mention the conditions of reversibility of a process (4)
  - (b) State and prove the Carnot's theorem (3)
  - (c) What is the efficiency of a Carnot engine operating between melting point and boiling Point of water under normal conditions. (7)
3.
  - (a) State and Explain the Biot and Savart law. Using it, deduce an expression for the magnetic Induction along the axis of a circular current carrying coil. (10)
  - (b) What are Maxwell's equations and explain their signature (4)
4.
  - (a) Define interference phenomena of light. (2)
  - (b) Deduce the conditions for maxima and minima of monochromatic light reflected from a Thin transparent (8)
  - (c) Describe the characteristics of lasers. (4)
5.
  - (a) Define numerical aperture of an optical fiber and what is its physical significance. (2)
  - (b) Deduce an expression for the numerical aperture of a fiber (7)
  - (c) Mention the important applications of ultrasonics (5)
6.
  - (a) What are matter waves and describe their properties . (3)
  - (b) Deduce the Schrodingers time independent wave equation. (8)

- (c) Give a classification of materials based on the band theory of solids (3)
7. (a) What are nano materials and describe the methods of characterizing the nano materials (6)  
(b) Describe with neat figure, any one method of synthesis the nano materials. (8)
8. Write about
- (a) Entropy and disorder (4)  
(b) Requirement of any laser device (4)  
(c) Piezoelectric method of producing ultrasonics (6)

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**[B16 ENG 1202]**

**[B16 ENG 1203]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**CHEMISTRY**  
MODEL QUESTION PAPER  
(Common to ECE, EEE, Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) What is hardness of water.
  - b) How solids are classified?
  - c) What are ceramics?
  - d) What is the Galvanic corrosion?
  - e) What do you mean by conducting polymers?
  - f) Define cetane number?
  - g) Write any two advantages of LPG as motor fuel.
2. (a) Describe the ion exchange process of water softening  
(c) Describe the steps involved in municipal water treatment.
3. (a) Write the manufacture of Portland cement.  
(b) Write the properties and applications of ceramics.
4. (a) Give a detailed account on band theory of solids.  
(b) Explain zone refining of solids with neat diagram.
5. (a) What is corrosion ? Explain the theory of dry corrosion with examples.  
(b) What is paint? Explain the constituents of paint.
6. (a) Define polymerization. Explain the mechanism of addition polymersation with suitable Examples.  
(b) Write the preparation and properties of cellulose derivatives
7. (a) Describe the manufacture of coke by Otto- Hoffmann's process  
(b) What is synthetic petrol? Explain Fischer Tropsch, method with a neat diagram.
8. (a) Explain the desalination of water by reverse osmosis method.  
(b) Explain the principles of Lubrication with neat diagram.

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**[B16 ENG 1203]**

**[B16 ENG 1204]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ENGINEERING GRAPHICS**  
**MODEL QUESTION PAPER**  
(Common to CIVIL, CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) What is an involute? Write its uses?
  - b) Define HT and VT.
  - c) What is an auxiliary plane? State its purpose?
  - d) Define frustum of a solid.
  - e) Define the term section plane.
  - f) State methods of developments.
  - g) Define isometric scale.
2. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight.
3. A line AB, of 80 mm long has its end A, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP.
4. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground.
5. A pentagonal pyramid, with base 30mm and height 80mm, rests on one edge of its base on HP. The highest point in the base is 30mm above HP. Draw its projections, when the axis is parallel to VP. Drawn another front view, on a reference line inclined at  $30^0$  to the edge on which it is resting, so that the base is visible.
6. A cone, base 75 mm diameter and axis 80 mm long is resting on its base on the H.P. it is cut by a section plane perpendicular to the V.P., inclined at  $45^0$  to the H.P. and cutting the axis at a point 35 mm from the apex. Draw its front view, sectional top view and true shape of the section.
7. A right regular hexagonal pyramid of 30 mm side of base and height of 70 mm stands with its base on HP. A through circular hole of 30 mm diameter is drilled through the pyramid such that

the axis of the hole is perpendicular to VP and intersects the axis of the pyramid 20 above the base. Draw the development of the lateral surface of the pyramid showing the true shape of the holes formed on it.

8. A right circular cylinder 5cm diameter of base and 7cm height has its base in the HP. A right circular cone diameter of base 4cm and height 4cm rests centrally over the upper flat surface of the cylinder. Draw the isometric view of the above combination.

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**[B16 ENG 1204]**



**[B16 ENG 1205]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**COMPUTER PROGRAMMING USING C & NUMERICAL METHODS**  
MODEL QUESTION PAPER  
(Common to ECE, EEE & Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

- 1 Write a short answer to the following. [ 7x2 = 14 marks]
- a) What is recursion? Give an example.
  - b) Explain scope and extent of variables.
  - c) What are truncation and round off errors?
  - d) Distinguish between local and global variables.
  - e) Explain different bitwise operators?
  - f) Explain Euler's Method.
  - g) What is Interpolation?
- 2
- a) Explain different types of operators in C.
  - b) Write a program to check whether the given number is palindrome or not.
- 3
- a) What is an array? Explain two dimensional array with an example?
  - b) Write a C program to generate prime numbers less than the given number.
- 4
- a) What is a Pointer? How is it initialized? What is the function of a pointer variable? What are its uses?
  - b) What is a loop ?Explain general forms of all loop structures with suitable examples.
- 5
- a) Explain the difference between call by reference & call by value?
  - b) Write a program to sort an array of elements in ascending order?

6 a) Explain the following

i) Structure

ii) Accessing elements in structure

iii) Arrays of structures

b) Briefly explain file handling functions.

7 a) Use gauss elimination method to solve

$$2x+y+z=10, 3x+2y+3z=18, x+4y+9z=16$$

b) Given  $y' = y - x$ , where  $y(0)=2$  find  $y(0.1)$  and  $y(0.2)$  using Runge-kutta fourth order method

8 a) Find the root of the following equation using Newton-Raphson method, correct the result upto 3 decimal places.

$$X^3 - 3X - 5 = 0.$$

b) Evaluate

$$\int_{-2}^2 x \sin(x) dx \text{ using Simpson's rule.}$$

$$-2$$

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[B16 ENG 1205]

**[B16 ENG 1206]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**PROFESSIONAL ETHICS AND MORAL VALUES**  
MODEL QUESTION PAPER  
(Common to CIVIL, CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - (a) Ethical Vision
  - (b) Profession and Professionalism
  - (c) Environmental Ethics
  - (d) Bhopal Gas Tragedy
  - (e) Gender discrimination
  - (f) Cyber Crimes
  - (g) Engineers as Managers
2. Discuss the scope and aim of Engineering Ethics.
3. Explain the role of Engineers in promoting ethical climate.
4. What are Values? Explain in detail the classification of human values.
5. Elucidate the moral responsibility of engineers towards safety and risk.
6. Define the concept of globalization and explain the role of MNCs in our country.
7. What are the functions of various sample codes of ethics?
8. Discuss the need to focus on professional ethics.

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**[B16 ENG 1206]**

**[B16 ENG 1207]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**HISTORY OF SCIENCE AND TECHNOLOGY**  
MODEL QUESTION PAPER  
(Common to ECE, EEE, Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) Explain the terms Science and Technology.
  - b) Describe the role of Scientist in the society.
  - c) Science and Technology Policy resolutions.
  - d) Defense Spin-offs.
  - e) Biosensors.
  - f) Barriers of Technological change.
  - g) Types of Technology transfer.
2. Describe the roots of science and technology in ancient period in India.
3. Explain the salient features of new technology fund and programs aimed at technological self reliance.
4. Describe the achievements of Council of Scientific and Industrial Research.
5. Explain the salient features of Space program and INSAT services.
6. Explain the importance of Nuclear energy and describe the nuclear explosion and India's safety measures.
7. Describe the importance of Ocean development and explain the marine research and capacity building.
8. What is Appropriate technology? Explain the criteria for selection of an appropriate technology.

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**[B16 ENG 1207]**

**[B16 CE 1208]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**BUILDING MATERIALS AND BUILDING CONSTRUCTION**  
MODEL QUESTION PAPER  
(Department Subject - CIVIL)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
  - a) Differentiate between wet and dry process manufacturing of ordinary Portland cement.(OPC)
  - b) How do you diagnose defects in painting suggest remedies
  - c) What is bearing capacity of soil? What is its importance?
  - d) Draw neat sketch of dog-legged staircase and quarter landing staircase
  - e) what are differences between distemper and emulsion paint
  - f) Explain decay of timbers
  - g) Define scaffolding and mention its components parts
2.
  - a) Discuss various methods of storing cement in the field and in godowns
  - b) Define Form Work and explain the different types of form-work.
3.
  - a) Draw the cross section of a tree and indicate various details.
  - b) Explain the properties of glass. What are the uses of glass brick and sheet glass?
4.
  - a) Describe various types of Pile foundations with brief description and usual notations
  - b) Explain about concrete blocks and FAL-G blocks
5.
  - a) How concrete is graded as per I.S.code? List out the factors effecting choice of mix problems.
  - b) Draw the sketch of queen post truss with all details
6.
  - a) Discuss the importance of location of doors, windows and ventilators in a building.
  - b) Explain the chemistry of plastics. Enumerate the various uses of plastics in buildings.
7.
  - a) Bring out the importance of aluminum and PVC doors, Windows and ventilators in building construction.
  - b) Describe the constituents of varnishes and explain the uses of varnishes
8.
  - a) What is a step? Mention its different types
  - b) Write short note on Transporting, placing and vibrating of concrete.

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**[B16 CE 1208]**

**[B16 CS 1208]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**PROBABILITY, STATISTICS & QUEUING THEORY**  
MODEL QUESTION PAPER  
(Department Subject-Common to CSE, IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

- 1 Write a short answer to the following [ 7x2 = 14 marks]
  - (a) State the limitations of axiomatic approach to probability.
  - (b) State the properties of distribution function.
  - (c) Show that  $E(aX + b) = aE(X) + b$ .
  - (d) Find the moment generating function of Poisson distribution
  - (e) Define rank correlation?
  - (f) Define Type-I-error and Type-II-error.
  - (g) What are the operating characteristics of a queuing model?
- 2
  - (a) State and prove addition theorem of probability for n events.
  - (b) Three machines A, B and C produce respectively 50%, 30% and 20% of the total number of items of a factory. The percentage of defective output of these machines is 3%, 4% and 5%
    - (i) If an item is selected at random, find the probability that the item is defective.
    - (ii) Suppose an item is selected at random and is found to be defective. Find the probability that it was produced by machine A.
- 3
  - (a) The diameter of an electric cable is assumed to be a continuous variate with p.d.f.  $f(x) = 6x(1-x)$ ,  $0 \leq x \leq 1$ . Verify that the above is p.d.f. Also find the mean and variance.
  - (b) Let X be a random variable with the following probability distribution:

|               |     |     |     |
|---------------|-----|-----|-----|
| x :           | -3  | 6   | 9   |
| P ( X = x ) : | 1/6 | 1/2 | 1/3 |

Find  $E(X)$ ,  $E(X^2)$  and using the laws of expectation, evaluate  $E(2X+1)^2$
- 4
  - (a) Twenty identical coins each with probability  $P$  of showing heads are tossed. The probability of heads showing on 10 coins is same as that of heads showing on 11 coins. Find  $P$ .
  - (b) X is a normal variate with mean 30 and standard deviation 5. Find the probability that
    - (i)  $26 \leq X \leq 40$
    - (ii)  $X \geq 45$
    - (iii)  $|X - 30| > 5$

- 5 (a) Obtain the equations of two lines of regression for the following data. Also obtain the estimate of  $X$  for  $Y = 70$

|   |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|
| X | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
| Y | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |

- (b) Find the correlation coefficient for the following data:

|     |    |    |    |    |    |    |    |    |    |    |
|-----|----|----|----|----|----|----|----|----|----|----|
| x : | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| Y : | 10 | 12 | 16 | 28 | 25 | 36 | 41 | 49 | 40 | 50 |

- 6 (a) A sample of 100 items, drawn from a universe with mean value 64 and standard deviation 3 has a mean value 63.5. Is the difference in means significant? What will be your inference if the sample has 200 items?

- (b) Determine a 95% confidence interval for the mean of a normal population with the Sample 145, 146, 142, 143

- 7 (a) A group of 10 boys fed on a diet A and another group of 8 boys fed on a different diet B recorded the following increase in weights.

|        |   |   |   |   |    |   |   |   |   |    |     |
|--------|---|---|---|---|----|---|---|---|---|----|-----|
| Diet A | 5 | 6 | 8 | 1 | 12 | 4 | 3 | 9 | 6 | 10 | Kgs |
| Diet B | 2 | 3 | 6 | 8 | 10 | 1 | 2 | 8 |   |    | Kgs |

Does it show the superiority of Diet A over that of Diet B

- (b) Theory predicts that the proportion of beans in four groups A, B, C, D should be 9:3:3:1. In an experiment among 1600 beans, the numbers in the four groups were 882, 313, 287, 118. Does the experiment support the theory?

- 8 (a) For  $\{(M/M/1):(\infty/FIFO)\}$  queuing model, in the steady state case, obtain the average queue length in terms of relevant parameters  $\lambda$  and  $\mu$ .

- (b) Arrivals at a telephone booth are considered to be Poisson with an average time of 12 min. between one arrival and the next. The length of phone call is assumed to be distributed exponentially with mean 4 min.

(a) Find the average number of persons waiting in the system.

(b) What is the probability that a person arriving at the booth will have to wait in the Queue?

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[B16 CS 1208]

**[B16 EC 1208]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ELECTRONIC DEVICES AND CIRCUITS**  
MODEL QUESTION PAPER  
(Department Subject-ECE)

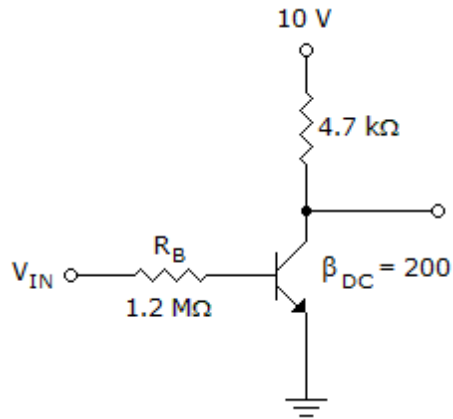
**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following [ 7x2 = 14 marks]
  - a. What is meant by diffusion current in a semi-conductor?
  - b. A silicon diode has a saturation current of 7.5 pA at 300 °K. Calculate the saturation current at 330 ° K.
  - c. Define peak inverse voltage of a diode.
  - d. Draw the input and output characteristics of a transistor in CE configuration and mark the cutoff, saturation and active regions.
  - e. Compare JFET with BJT.
  - f. Define pinch-off voltage.
  - g. Draw the equivalent circuit of transistor for high frequencies
2.
  - a. Explain the current components in a PN junction diode and Derive the diode current equation.
  - b. Explain about avalanche and zener breakdowns.
3.
  - a. Explain about intrinsic and extrinsic semiconductors
  - b. write short note on (i) Hall effect (ii) continuity equation
4.
  - a. Explain the working of Bridge rectifier. Give the expressions for RMS current, PIV, ripple factor and efficiency.
  - b. A diode whose internal resistance is  $20\Omega$  is to supply power to a  $100\Omega$  load from 110V(rms) source supply. Calculate (i) peak load current (ii) the dc load current (iii) the ac load current (iv) the percentage regulation from no load to full load.
5.
  - a. Draw and explain the input and output characteristics of a transistor in CB configuration.
  - b. Determine the minimum value of  $I_B$  that produces saturation in the following figure.





6. a. Explain with the help of neat diagrams, the structure of an N-channel FET and its Volt-ampere characteristics.  
 b. Explain the operating principle of enhancement mode MOSFET. How does it differ from depletion mode type?
7. a. Explain how FET acts as a voltage variable resistor.  
 b. Show that if a FET is operated at sufficiently low drain voltage, it behaves as a resistance  $R$  given by  $R = R_O / [1 - (V_{GS} / V_P)^{1/2}]$  Where  $R_O$  is the channel resistance for zero gate voltage.
8. Write a short notes on
  - a. Photo transistor
  - b. Tunnel diode
  - c. Transition capacitance

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[B16 EC 1208]

**[B16 EE 1208]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**CIRCUIT THEORY**  
MODEL QUESTION PAPER  
(Department Subject-EEE)

**Time: 3 Hrs.**

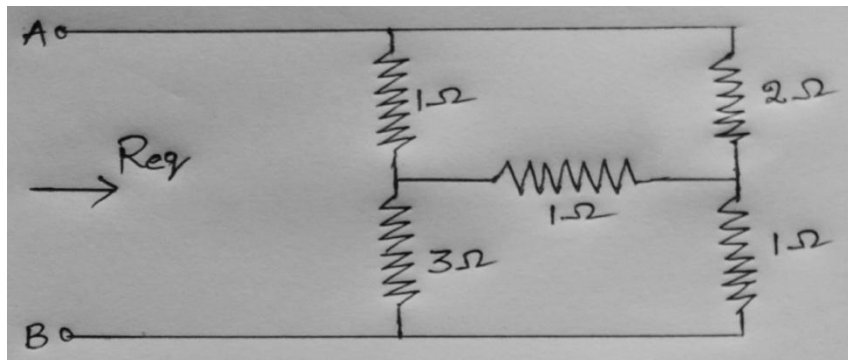
**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

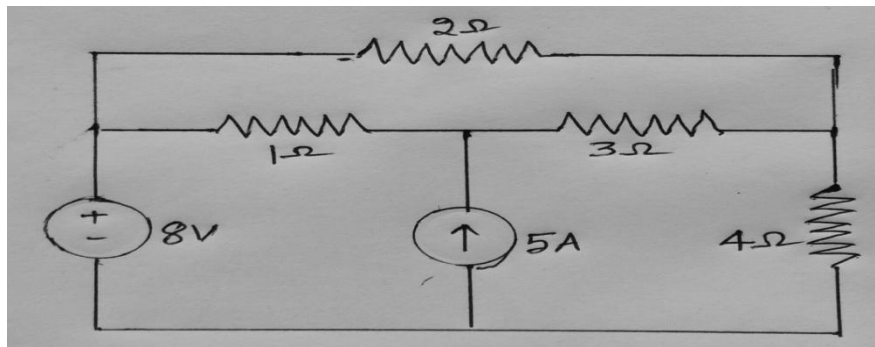
1. Write a short answer to the following [ 7x2 = 14 marks]

- a) What are the limitations of superposition theorem?
- b) Distinguish between Active and Passive elements.
- c) Draw the characteristics of an ideal voltage source.
- d) State Maximum power transfer theorem.
- e) Define MMF, Reluctance and Magnetic flux with respect to a magnetic circuit.
- f) State Faradays laws of Electromagnetic Induction.
- g) What is Self inductance?

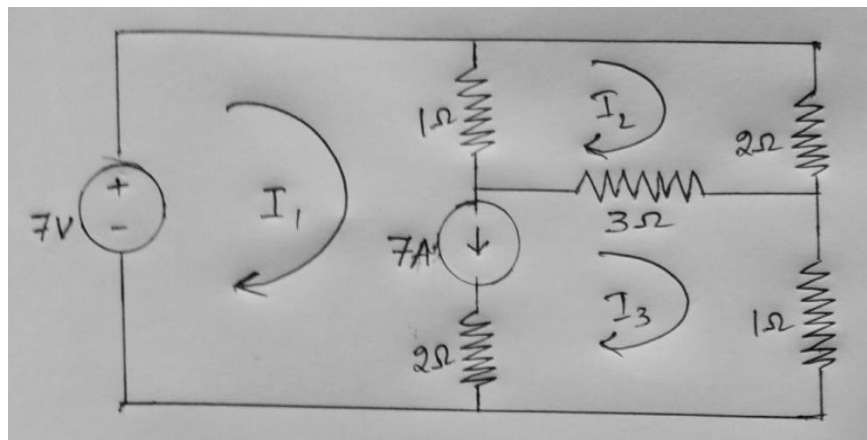
2. a) Find the equivalent resistance between the terminals A and B of the given network.



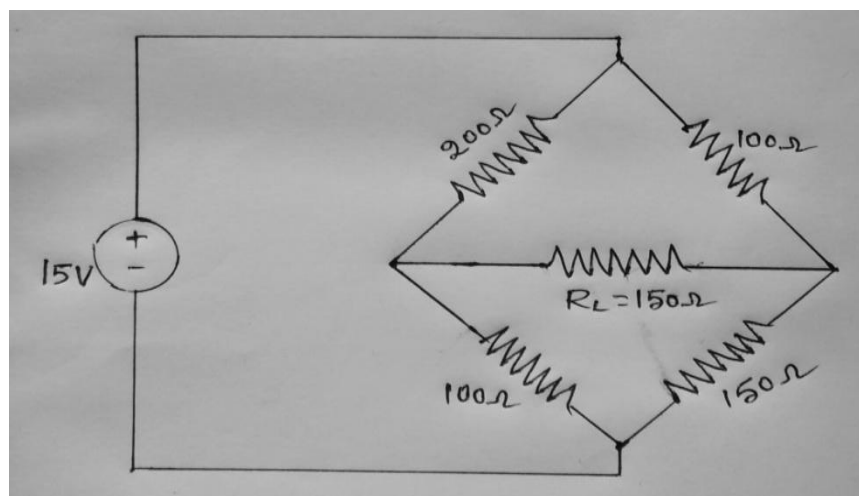
b) Using Nodal analysis find the currents and voltages in all the branches of the given network.



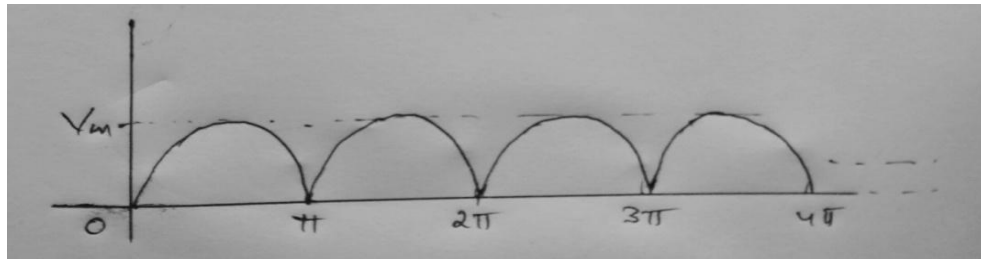
3.a) Find the mesh current  $I_1$  in the given circuit using mesh Analysis.



b) Find the current through the load resistance  $R_L$  in the given circuit using thevenin's theorem.



4. a) Define Average, RMS values of a periodic waveform. Obtain the Average and RMS values of the rectified sinusoidal waveform shown in figure.



- b) A series R-L-C circuit has  $R=10\Omega$ ,  $L=0.01\text{H}$ ,  $C=100\mu\text{F}$ . Find the Resonant frequency, Quality factor and Band width of the circuit.
- 5.a) Explain (i) Instantaneous power (ii) Average power (iii) Complex power applied to AC circuits.
- b) The supply voltage to a circuit is  $v(t)=220\sqrt{2}\sin(\omega t)$  and the current drawn from it is  $i(t)=14.14\sin(\omega t - 45^\circ)$ . Find the Apparent, Active and Reactive powers.
6. a) Obtain the relation between Line and Phase quantities in a Star connected circuit.
- b) A 220V, 3-phase voltage is applied to a balanced delta connected 3-phase load of  $(15+j20)\Omega$  ohms per phase. Find (i) Phasor current in each line (ii) Power consumed per phase and (iii) Phasor sum of three line currents and comment on it.
7. a) Distinguish between Statically induced emf and Dynamically induced emf.
- b) An iron ring of cross-sectional area of  $10\text{ cm}^2$  is wound with a wire of 1500 turns has a saw cut of 3mm air gap. Calculate the magnetizing current required to produce a flux of 0.25mwb if the mean length of the magnetic path is 50cm and relative permeability of 470 and the leakage factor is 1.2.
8. a) What are the advantages of three phase circuits? (4M)
- b) Give the Analogy between Electric and Magnetic circuits. (4M)
- c) Explain about the measurement of power in three phase circuits. (6M)

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[B16 EE 1208]

**[B16 ME 1208]**  
I/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**METALLURGY AND MATERIALS ENGINEERING**  
MODEL QUESTION PAPER  
(Department Subject-Mechanical)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsorily.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short answer to the following. [ 7x2 = 14 marks]
- a) Define lattice parameters.
  - b) Define Gibbs phase rule
  - c) Explain peritectic transformation
  - d) Define heat treatment
  - e) Write a short note on isothermal transformation curves
  - f) Define smart materials.
  - g) Write short notes on fiber composites.
2.
  - a) Discuss various types of defects in crystals?
  - b) Explain different crystal structures and find the atomic packing factor for BCC, and FCC structures.
3.
  - a) With a neat sketch explain iron-carbon phase diagram and label all its phases.
  - b) What is a phase diagram? And discuss the construction of phase diagrams.
4.
  - a) What are the different steps to construct isothermal transformation curves for a eutectoid steel and explain it.
  - b) Explain the Austempering and Martempering process.
5.
  - a) Define composite materials? Discuss briefly various reinforcements in composite materials.
  - b) Mention advantages, limitations and applications of particle- reinforced composites.
6.
  - a) Explain the composition and application of the following.  
i) Hadfield Steels, ii) Tool Steels, iii) High Speed Steels
  - b) What are different types of cast irons and explain how malleable cast iron is produced.
7.
  - a) What are the different case hardening methods and explain Carburizing process.
  - b) Explain flame and Induction hardening process with neat diagram.

- 8 Write a short note on any THREE of the following
- a) Nano materials
  - b) Invariant reactions
  - c) Applications of composites
  - d) Concept of Slip and Twinning
  - e) Precipitation Hardening

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**[B16 ME 1208]**

**[B16 ENG 2101]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester.  
**MATHEMATICS-IV**  
MODEL QUESTION PAPER  
(Common to CIV, ECE, EEE & ME)

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

- 1 (a) Find a unit vector normal to the surface  $x^3 + y^3 + 3xyz = 3$  at the point  $(1, 2, -1)$ .  
(b) Show that  $\text{Curl}(\text{grad } \phi) = 0$ .  
(c) Show that  $\iiint_S \nabla r^2 \cdot d\vec{s} = 6V$ .  
(d) State two-dimensional Laplace equation in Cartesian coordinates. Define harmonic function.  
(e) Find the analytic function whose real part is  $x^3 - 3xy^2$ .  
(f) Evaluate  $\oint_C \frac{z^2 - z + 1}{(z - 2)} dz$  where C is the circle  $|z| = 1$ .  
(g) Find the nature and location of the singularities of the function  $\frac{1}{(z - 1)^3}$ .
- 2 (a) Find the directional derivative of  $f = x^2 - y^2 + 2z^2$  at the point  $P(1, 2, 3)$  in the direction of the line PQ where Q is the point  $(5, 0, 4)$ . Also calculate the magnitude of the maximum directional derivative.  
(b) Prove that  $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$ .
- 3(a) Show that  $\vec{F} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k}$  is a conservative field. Find the potential function and hence the work done in moving a particle in this field from  $(1, -2, 1)$  to  $(3, 1, 4)$ .  
(b) Use Green's theorem to evaluate  $\oint_C [(3x - 8y^2)dx + (4y - 6xy)dy]$  where C is the boundary of the region described by  $x = 0, y = 0$  and  $x + y = 1$
- 4 (a) Evaluate  $\iiint_V \text{div } \vec{F} dv$  where  $\vec{F} = y\vec{i} + x\vec{j} + z^2\vec{k}$  over the cylindrical region bounded by  $x^2 + y^2 = 9, z = 0$  and  $z = 2$ .  
(b) Find components of the vector field  $z\vec{i} - z\vec{j} + y\vec{k}$  in cylindrical polar coordinates.
- 5 (a) Solve the equation  $py^3 + qx^2 = 0$  by the method of separation of variables.

(b) A tightly stretched string with fixed end points  $x = 0$  and  $x = l$  is initially at rest in its equilibrium position. If it is set vibrating by giving to each of its points a velocity  $\lambda x(l-x)$ , find the displacement of the string at any distance  $x$  from one end at any time  $t$ .

6 (a) If  $f(z)$  is a regular function of  $z$ , prove that  $\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$ .

(b) Find the bi-linear transformation which maps the points  $z = 1, -1, \infty$  of the  $z$ -plane onto the points  $w = 1+i, 1-i, 1$  of the  $w$ -plane. Hence find the critical points and the invariant points of this transformation.

7(a) Evaluate  $\oint_C \frac{z^3 + z + 1}{z^2 - 7z + 2} dz$ , where  $C$  is the ellipse  $4x^2 + 9y^2 = 1$ .

(b) Find the Laurent's expansion of the function  $f(z) = \frac{1}{(1-z)(2-z)}$  valid for

(i)  $0 < |z-2| < 1$       (ii)  $|z-1| > 1$ .

8 (a) Evaluate  $\int_C \tan z \, dz$  where  $C$  is the circle  $|z| = 2$ .

(b) Use calculus of residues to evaluate  $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos \theta} d\theta$ .



**[B16 CE 2101]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**ENGINEERING MECHANICS**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

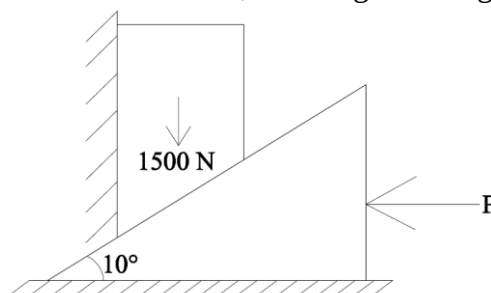
**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

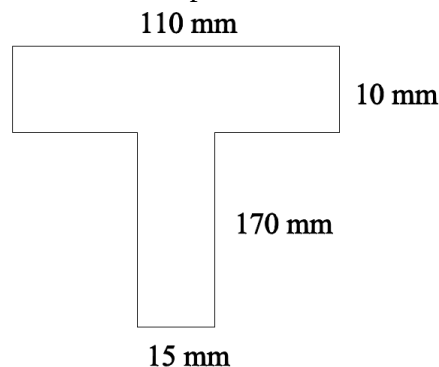
- 1 a) state Varignon's theorem  
b) What is the difference between collinear and concurrent forces?  
c) What is product of inertia?  
d) State theorem of Pappus?  
e) Explain law of conservation of energy.  
f) Define velocity of projection, angle of projection and time of flight.  
g) What is direct impact?

2 a) state and prove parallel axis theorem.

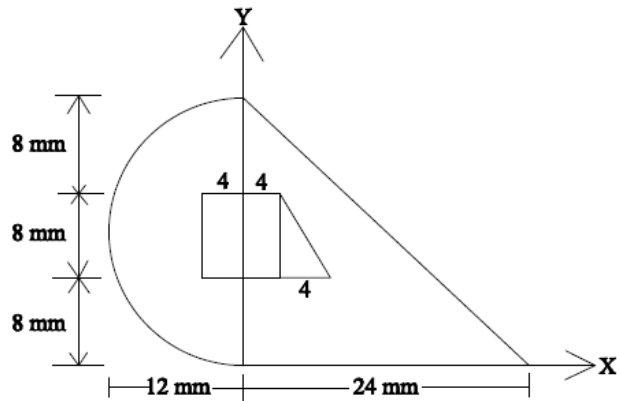
b) A block over lying a  $10^\circ$  wedge on a horizontal floor and leaning against a vertical wall and weighing 1500N is to be raised by applying a horizontal force to the wedge. Assuming co-efficient of friction between all the surfaces in contact to be 0.3, determine the min horizontal force to be applied to raise the block, referring to the figure below.



3 a) Determine the moment of inertia for the plane area about its centroidal x-axes.

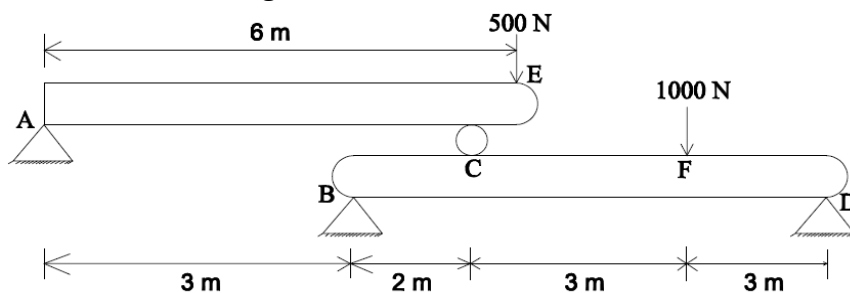


b) Determine the Centroid of the shaded area as shown in fig with respect to x and y axes. All dimensions are in mm.

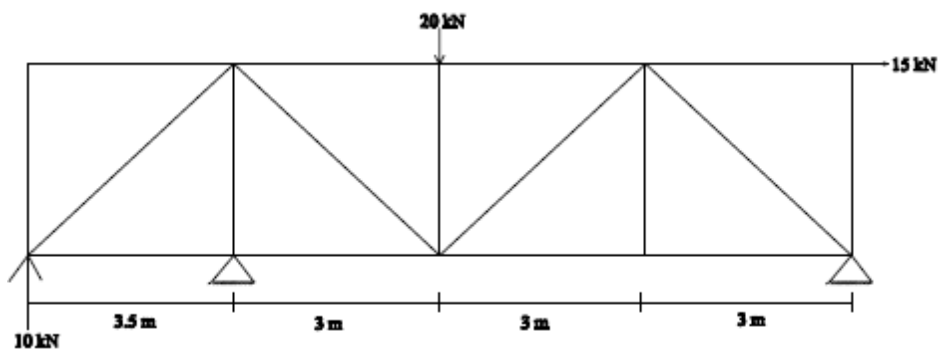


4 a) State and explain principle of virtual work.

b) Using the principle of virtual work, find the reactions at rollers 'C' and 'B' of the assembly loaded as shown in fig.



5) Determine the forces in the members of the truss shown in fig. Height of the truss 4m.



6 a) a bullet is fired at 125m/s from a point A and has to hit a target at a horizontal distance of 1000m from A and situated 200m higher than A. Find

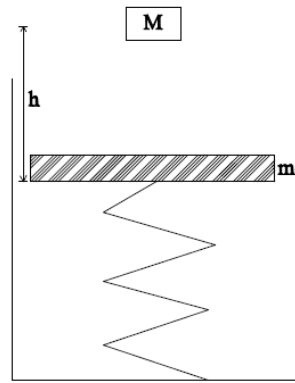
i) The angle  $\alpha$  at which the bullet must be fired in order to strike the target in the minimum time,

ii) The maximum height above A reached by the bullet.

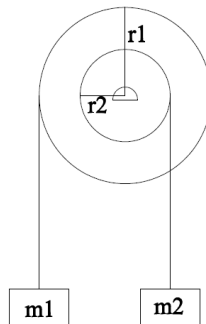
iii) The time of flight.

b) a body of mass 8kg moving with a velocity of 10m/s collides with another body of mass 3 kg moving with 5m/s in opposite direction. Find the velocities of the individual bodies after impact if the coefficient of restriction is 0.9

7 a) A block weighing  $M=27\text{Kg}$  is dropped on the pan of a spring scale, shown in fig. The height of drop  $h=1.7\text{m}$  and weight of scale pan  $m=7\text{kg}$ . assuming a perfectly plastic impact, determine the max amount of deflection of the spring where spring constant  $K=20\text{N/mm}$ .



b) Consider the inertial and the frictional effect absent in the pulley system . Determine the downward acceleration of mass  $m_1$  as shown in fig take  $m_1=8\text{kg}$ ,  $m_2=12\text{kg}$ ,  $r_1=2r_2=20\text{cm}$ .



8a) State and prove work energy principle.

b) What is instantaneous centre of rotation? Show that for a ladder sliding along two mutually perpendicular walls, the instantaneous centre takes the path of a circle.

**[B16 CE 2102]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**MECHANICS OF SOLIDS**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks:70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

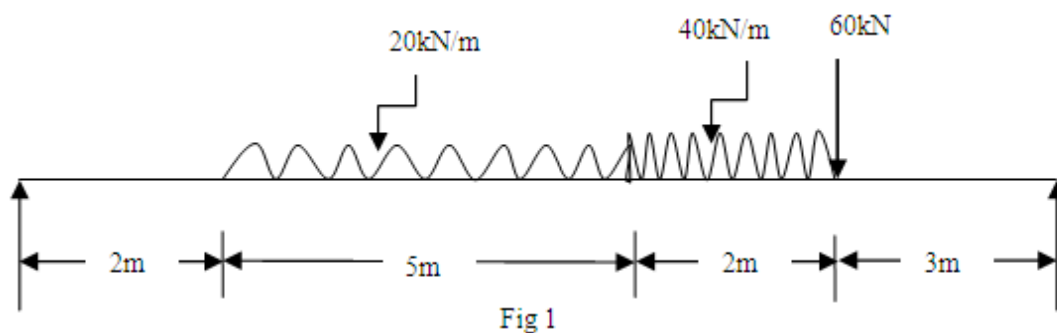
1.

- (a) State Hook's law with suitable example.
- (b) Define Poisson's ratio and factor of safety.
- (c) Discuss the sign conventions for shear force and bending moments used in beams.
- (d) Define point of contraflexure.
- (e) List the assumptions made in the theory of simple bending.
- (f) Define the term pure torsion with suitable examples.
- (g) What is meant by crippling load and slenderness ratio?

2. (a) Draw the stress- strain diagram for Mild Steel and discuss the salient features.

(b) A rod of steel is 5 m long and 30mm diameter at a temperature of 28°C. Find the free expansion of the rod when temperature is raised to 75°. Determine the stress and pull exerted if the ends do not yield.  $E = 2 \times 10^5 \text{ N/mm}^2$ ,  $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$ .

3. A simply supported beam has a span of 10 m and carry the loadings shown in fig 1. Draw S.F and B.M diagrams indicating salient values. Find the position and magnitude of the maximum BM in the beam.



4. (a) Find an expression for section modulus for a circular and hollow circular sections.

(b) Derive the relation  $\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$  for a beam

5. (a) A simply supported beam of span L is subjected to equal loads W/2 each at 1/3<sup>rd</sup> span points. Find the expressions for deflection under the load and at the mid span. Use Macaulay's method

(b) Distinguish between Double integration method and Macaulay's method

6. (a) Derive an expression for the major and minor principal stresses on an oblique plane, when the body is subjected to direct stresses in two mutually perpendicular directions accompanied by a shear stress

(b) At a point within a body subjected to two mutually perpendicular directions, the stresses are  $150 \text{ N/mm}^2$  (tensile) and  $100 \text{ N/mm}^2$  (compression). Each of the above stresses, is accompanied by a shear stress of  $100 \text{ N/mm}^2$ . Determine the normal, shear and resultant stresses on an oblique plane inclined at an angle of  $45^\circ$  with the axis of minor tensile stress.

7. (a) Find the maximum power transmitted by a shaft at 200 rpm without exceeding the permissible stress of 100 MPa if the shaft is (i) a solid circular shaft of diameter 60mm and (ii) a hollow shaft of the same internal diameter and has the same weight as the solid shaft.

(b) A closed coil helical spring is made with 12mm diameter wire and is having mean diameter of 150mm and 10 complete turns. The modulus of rigidity of the material of spring is  $80 \text{ kN/mm}^2$ . Find (i) Maximum shear stress, (ii) Strain energy stored, (iii) Deflection produced, and (iv) Stiffness of the spring.

8. A hollow cast iron whose outside diameter is 200mm and has a thickness of 20mm is 4.5m long and is fixed at both ends. Calculate the safe load by Rankine's formulae using a factor of safety of 2.5. Find the ratio of Euler's to Rankine's loads. Take  $E = 1 \times 10^5 \text{ N/mm}^2$  and Rankine's constant =  $1/1600$  case and  $f_c = 550 \text{ N/mm}^2$ .

[B16 CE 2102]

**[B16 CE 2103]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**FLUID MECHANICS –I**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1.
  - a. Distinguish real fluid – ideal fluid
  - b. State the difference in working principle of Manometers and Mechanical gauges.
  - c. Differentiate among stable, unstable and neutral equilibrium.
  - d. Define Steady flow and Non uniform flow.
  - e. State impulse momentum equation. what is its advantage over Bernoulli's equation?
  - f. Differentiate between Small and Large orifice .
  - g. List the various losses in flow through pipes.
2.
  - a. What are the physical properties of a fluid? Explain their practical significance.
  - b. An oil of viscosity 5.0 poise is used for lubrication between a shaft and sleeve. The diameter of the shaft is 0.5 m and it rotates at 200 rpm. Calculate the power lost in oil for a sleeve length of 100mm. The thickness of the oil film is 1.0 mm.
3. (a) How do you determine the hydrostatic force acting on a submerged curved surface?  
(b) A circular plate 1.5 m diameter is submerged in, water, with its greatest and least depths below the surface being 2 m and 0.75 m respectively. Determine
  - (i) The total pressure on one face of the plate and
  - (ii) The position of the centre of pressure
4. (a) Differentiate between convective and local acceleration.  
(b) The streamlines are represented by
  - (i)  $\psi = x^2 - y^2$
  - (ii)  $\psi = x^2 + y^2$
  - (1) Determine the velocity and its direction at (2,2).
  - (2) Sketch the streamlines and show the direction of flow in each case
5. (a) Derive Bernoulli's equation for flow along a stream line. Explain the practical application of Bernoulli's equation.

- (b) A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm is employed in horizontal pipe to measure the flow of water. The reading of the differential manometer connected to the inlet is 180 mm of mercury. If the coefficient of discharge is 0.98, determine the rate of flow.
6. (a) What are the advantages of a triangular notch over a rectangular notch? What is velocity of approach?
- (b) A 3m high tank Standing on the ground is kept full of water. There is a small orifice in its vertical side with its centre at depth  $h$  meters below the free surface of liquid in the tank. Find the value of  $h$  so that liquid strikes the ground at the maximum distance from the tank. Assuming  $a = 0.97$ , calculate the maximum value of the horizontal distance.
7. (a) Derive an expression for the force exerted by a Jet water on moving inclined plate in the direction of Jet.
- (b) In a  $45^\circ$  bend a rectangular air duct of  $1 \text{ m}^2$  cross sectional area is gradually reduced to  $0.5 \text{ m}^2$  area. Find the magnitude and direction of force required to hold the duct in Position if the velocity of flow at  $1 \text{ m}^2$  section is  $10 \text{ m/s}$  and pressure is  $30 \text{ KN/m}^2$ . Take the specific weight of air as  $0.0116 \text{ KN/m}^3$ .
8. (a) Explain Hardy-Cross method of analysis.
- (b) A pipeline of 600 mm diameter is I. long. To increase the discharge another of the same diameter is introduced parallel to the first in the second half of the length If  $f = 0.01$  and head at inlet is 300 mm, calculate the increase in discharge.

**[B16 CE 2104]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**SURVEYING**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write short answers for the following: (1X14=14M)
  - a. What is the Dip of a magnetic needle in compass?
  - b. Distinguish between accuracy and precision.
  - c. Differentiate between base line and check line.
  - d. What is a well-conditioned triangle?
  - e. What is profile levelling?
  - f. What is meant by traverse surveying?
  - g. What is the use of Ceylon Ghat tracer?
  - h. What is the basic principle of surveying?
  - i. What do you mean by line of collimation?
  - j. The true bearing of a line AB is  $220^{\circ}45'$  what will be its magnetic bearing if the declination is  $8^{\circ}15'$ .
  - k. Distinguish between isogonic lines and agonic lines.
  - l. What is parallax? How can you eliminate it?
  - m. State the limitations of plane table survey.
  - n. What are the uses of a optical square?
2. A.) What are the various errors of chaining? Mention whether they are cumulative or compensating.  
B.) A steel tape 30 m long at a temperature of  $70^{\circ}\text{F}$  when lying horizontally on the ground. Its sectional area is  $0.8\text{ cm}^2$ , weight 1.5 kg and  $\alpha = 60 \times 10^{-7}/\text{F}$ . The tape is structured over three spans. Calculate the actual length under the following conditions. Temperature  $800\text{F}$ , Pull 5 kg. Take  $E = 2.108 \times 10^6\text{ kg/cm}^2$ .
3. A.) What is meant by local attraction? How are the observed bearings corrected for local attractions?  
B.) Calculate the interior angles of the following traverse and exercise the geometric check.

| Line | Fore Bearing     |
|------|------------------|
| AB   | $70^{\circ}30'$  |
| BC   | $132^{\circ}00'$ |
| CD   | $56^{\circ}00'$  |
| DE   | $215^{\circ}30'$ |
| EA   | $310^{\circ}30'$ |



4. A.) Explain the following terms with reference to a compass survey.

- i) Magnetic Meridian and True Meridian
- ii) Whole circle bearing and reduced bearing

B.) The following are the bearings taken in running a compass traverse.

| Line | Fore Bearing     | Back Bearing     |
|------|------------------|------------------|
| AB   | $66^{\circ}15'$  | $244^{\circ}00'$ |
| BC   | $129^{\circ}45'$ | $313^{\circ}00'$ |
| CD   | $218^{\circ}30'$ | $37^{\circ}30'$  |
| DA   | $300^{\circ}45'$ | $126^{\circ}45'$ |

Determine the corrected as well as true bearings if the magnetic declination was  $1^{\circ}40'$  E.

5. A.) List the typical characteristics of contours. Explain the various indirect methods of contouring.

B.) The following reciprocal levels were taken with one level.

| Instrument at | Reading on |       | Remarks                                |
|---------------|------------|-------|----------------------------------------|
|               | A          | B     |                                        |
| A             | 1.564      | 2.868 | Distance AB = 100m<br>RL at A = 190.85 |
| B             | 0.536      | 1.732 |                                        |

Determine:

- i.) True difference in elevation between A and B
- ii.) The RL of B
- iii.) The Collimation error.

6. A.) Enumerate the various methods of plane table. Discuss the methods of radiation and intersection in details

B.) Explain the construction and working of a planimeter.

7. A.) What is a contour gradient? What are the uses of contour maps?

B.) Explain the graphical method in plane table surveying.

8. Write short notes on any THREE:

- i.) Abney Level
- ii.) Check lines and Tie lines
- iii.) Gale's Traverse Table
- iv.) Conventional Signs.

**[B16 ENG 2103]**  
II/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**ENVIRONMENTAL STUDIES**  
MODEL QUESTION PAPER  
(Common to CIV,CSE & IT)

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write short answers for the following:

- (a) Give the objectives of Environmental Studies
- (b) Define ecosystem
- (c) What are hotspots?
- (d) What is soil erosion?
- (e) What is sustainable development?
- (f) State the practical benefits of watershed management
- (g) What is biomagnifications movement?

2. Write about structure and function of forest ecosystem

3. Give an account of the various energy resources of India and their merits and demerits.

4. Give the bio-geographical classification of India and add a brief note on threats to biodiversity

5. Explain causes, effects and control measures of water pollution

6. Write a critical account of the effect of population growth on environment.

7. Give an account of rain water harvesting and watershed management with suitable example

8. Write short notes:

- a) Conflicts of water
- b) Effect of modern agriculture
- c) Noise pollution
- d) Solid waste management

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**[B16 ENG 2103]**

**[B16 CE 2201]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ANALYSIS OF STRUCTURES**  
**MODEL QUESTION PAPER**  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

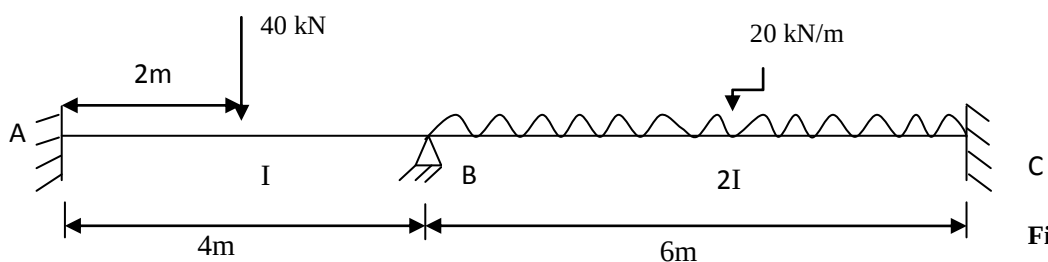
**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1.
  - a) Define a statically determinate structure
  - (b) State Castigliano's First theorem
  - (c) What is the degree of indeterminacy of a fixed beam- Substantiate your answer.
  - (d) Explain "Carry over Factor" with a sketch.
  - (e) Show the load positions for max +ve shear force and -ve shear force if a U.D.L shorter than the span is traversing from left to right of a beam.
  - (f) What are the advantages and disadvantages of moment distribution method?
  - (g) Differentiate between thin cylinder and thick cylinder.
2.
  - (a) A cantilever is loaded with a UDL throughout its length calculate the slope and deflection at the free end if the free end is propped to same level as that of the fixed end, calculate the prop reaction and the slope at the propped end. Use moment area method.
  - (b) A beam of constant cross section 4m long is free supported at its ends. It is loaded at points 1m from each end with a load of 2kN. Find the ratio of the deflections in the centre of the beam to the deflection at the point under one of the loads. Use conjugate beam method.
3.
  - (a) A fixed beam AB of length 6m is subjected to a constructed couple of 500 Nm(clockwise) at a point C which is at a distance of 3.5m from the end A. Draw SF & BM diagram assume "EI" to be constant.
  - (b) A beam AB 4m fixed at A & B carries a UDL of 1500N/m. The support B sinks by "1cm". Find the fixed end moments and draw BMD for the beam. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ ,  $I = 8000 \text{ cm}^4$ .
4. Analyse the two span continuous beam shown in figure (a) by slope deflection method draw bending moment and shear force diagram young's modulus is same throughout.

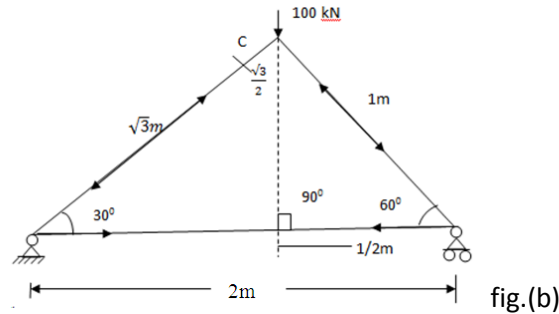


**Fig. (a)**

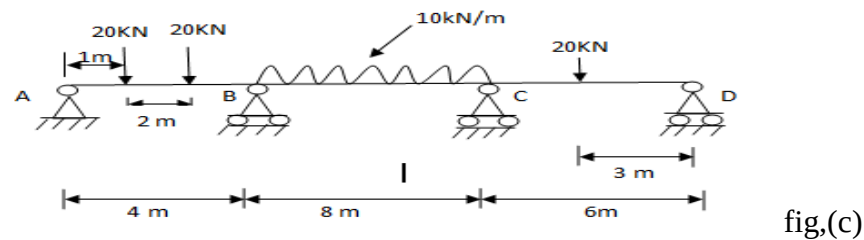
5. (a) A tension bar 5m is made up of two parts 3m of its length has cross sectional area of  $10\text{cm}^2$  while the remaining 2m has a cross sectional area of  $20\text{cm}^2$ . An axial load of 80 kN is gradually applied. Find the total strain energy produced in bar and compare this value with that obtained in a uniform bar of same length and having the same volume, when under the same load. Take  $E = 2 \times 10^5 \text{ N/mm}^2$ .

(b) Determine the vertical and horizontal displacements of joint C of the truss shown in fig.(b).

Given  $E = 200 \times 10^6 \text{ kN/m}^2$ ,  $A = 100 \times 10^{-6} \text{ m}^2$  for all members.



6. Analyse the continuous beam shown in fig.(c) using theorem of three moments.



7. The load system shown in fig. (d) crosses a simply supported beam over a span of 24 m. Determine (a) the maximum S.F. at a section 8 metres from the left hand end; (b) maximum B.M. under 25,000 N load.

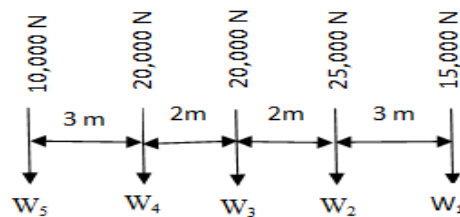


fig. (d)

8. (a) Explain different theories of failures.

(b) Find the thickness of the metal necessary for a cylindrical shell of internal diameter 160mm to with stand an internal pressure of  $8 \text{ N/mm}^2$ . The maximum hoop stress in the section is not to exceed  $35 \text{ N/mm}^2$ .

[B16 CE 2201]

**[B16 CE 2202]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**REINFORCED CONCRETE STRUCTURES**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**  
Use of IS 456:2000 is allowed

1. (a) What are the advantages of adopting limit state method of design over working stress method of design?  
(b) Mention the situations where the doubly reinforced sections are to be adopted.  
(c) Explain clearly the difference between flexural bond and development bond.  
(d) Distinguish between (i) short and long columns and (ii) unsupported length and effective length of a compression member.  
(e) Explain clearly the difference in the behavior of one way slab and Two way slabs.  
(f) How does the shear span influence the mode of shear failure?  
(h) Why is it necessary to provide transverse reinforcement in a one way slab?
2. (a) Estimate the concrete stress block parameters in compression in limit state method.  
(b) Determine the moment of resistance of a rectangular beam section 250 mm wide and 400 mm depth. When it is reinforced with 3Nos. of 22 mm diameter bars on compression side and 3Nos. of 28 mm bars on tension side both at an clear cover of 30 mm for M 25 concrete and Fe415steel.
3. A rectangular reinforced concrete beam of 7 m span (centre- to-centre of supports) resting on 300 mm wide simple supports, is to carry a uniformly distributed dead load (excluding self-weight) of 15 kN/m and live load of 20 kN/m. Design the beam section for maximum moment at midspan. Use M 25 grade concrete and Fe 415 steel.
4. A simply supported beam  $300 \times 600$  mm (effective) is reinforced with five bars of 25 mm diameter. It carries a characteristic load of 80 kN/m( including its own weight) over an effective span of 6 m. out of five main bars, two bars can be bentup safely near the support. Design the shear reinforcement for the beam. Using M 20 grade of concrete and Fe 415 steel.
5. Design a simply supported slab to cover a hall with internal dimensions  $4.0 \text{ m} \times 6.0 \text{ m}$ . The slab is supported on masonry walls 230 mm thick. Assume a live load of  $3 \text{ kN/m}^2$  and a finish load of  $1.0 \text{ kN/m}^2$ . Use M 20 concrete and Fe 415 steel. Assume that the slab to be prevented from lifting up.
6. Design the reinforcement for a column with  $l_{ex} = l_{ey} = 3.5 \text{ m}$  and size  $300 \times 500 \text{ mm}$ , subject to a factored axial load of 1250 kN with biaxial moments of 180 kNm, and

100 kNm (i.e.,  $M_{ux} = 180 \text{ kNm}$ ,  $M_{uy} = 100 \text{ kNm}$ ). Assume M 25 concrete and Fe 415 steel.

7. Design a square footing for a rectangular column 300 mm × 500 mm, reinforced with 6-25 mm  $\emptyset$  bars, and carrying a service load of 1250 kN. Assume soil with an allowable pressure of  $200 \text{ kN/m}^2$  at a depth of 1.25 m below ground. Assume Fe 415 grade steel for column and footing, and M 20 grade concrete for the footing and M 25 grade concrete for column.
8. Design the torsional reinforcement in a rectangular beam section, 350 mm wide and 750 mm deep, subjected to an ultimate twisting moment of 140 kNm, combined with an ultimate bending moment of 200 kNm and an ultimate shear force of 110 kN. Assume M 25 concrete, Fe 415 steel.

**[B16 CE 2203]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**FLUID MECHANICS II**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. (a) What is difference between Couette and Poiseuille flow?  
(b) State the characteristics of laminar boundary layer.  
(c) What is Von Karman Vortex trail?  
(d) What is meant by most economical channel section  
(e) Explain the functions of Parshall flume  
(f) List the various types of Hydraulic jump.  
(g) What is Canal transition?
  
- 2 (a) Prove that the average velocity is  $\frac{2}{3}$ <sup>rd</sup> of the maximum velocity for a steady laminar flow between two fixed parallel plates.  
  
(b) Two fixed parallel plates kept 8cm apart have Laminar flow of oil between them with a maximum velocity 1.5m/s. Taking dynamic viscosity of oil to be  $\mu = 2 \text{ N.s/m}^2$ . Compute :  
  - (i) the discharge per metre width
  - (ii) Shear stress at the plates
  - (iii) The pressure difference between the two points 25cm. apart
  - (iv) Velocity at 2 cm. from the plate
  - (v) The velocity gradient at the plates end.
  
3. (a) Explain the phenomena of boundary layer separation and methods of controlling it.  
(b) The velocity distribution in the boundary layer is given by  $\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta}\right) - \frac{1}{2} \left(\frac{y}{\delta}\right)^3$ , being the boundary layer thickness. Calculate  
  - (i) Displacement thickness and
  - (ii) Momentum Thickness
  
- 4 (a) Derive the condition for maximum discharge for a given value of specific energy  
  
(b) A kite of Surface area  $0.4 \text{ m}^2$  flies in the air making an angle of  $25^\circ$  with the horizontal. The weight of the kite is 1.5 N and string tension is 5N. The string makes an angle of  $70^\circ$  with plane of the kite. If the wind velocity is 20 kmph, find the drag and lift coefficients of the kite. Density of air =  $1.8 \text{ kg/m}^3$
  
- 5 (a) Derive the dynamic equation of gradually varied flow in open channel

(b) Water enters to a rectangular channel of uniform cross section at a velocity of 8m/sec. The depth of water is 60 cm. Verify if the flow is sub critical and calculate the critical depth and corresponding velocity. Also calculate the specific energy at the inlet and at critical condition.

6 (a) Draw a neat sketch of the specific energy curve for an open channel with constant discharge. Define critical flow in an open channel and derive general criteria for such flow

(b) A rectangular flume 2m wide carries discharges at the rate of  $2 \text{ m}^3/\text{s}$ . the bed slope of the flume is 0.0004. At a section the depth of flow is 1 m. Calculate the distance of the section downstream where the depth of the flow is 0.9 m. solve by single step method. Assume rugosity coefficient as 0.014. Is the slope of the channel mild or steel? How is this type of surface profile classified?

7 (a) Sketch the possible gradually varied flow profiles for the following arrangement of channels and controls

(i) Steep-horizontal mild ending with storage

(ii) A hydraulic jump occurs in a 0.5m. wide rectangular channel at the point, where depth of water flow is 0.15 m. and Froude number is 2.5. calculate the specific energy, critical and sequent depths, loss of head and energy dissipated.

8 write short notes on:

(a) Boussinesq eddy viscosity model

(b) Prandtl velocity profile

(c) Types of hydraulic jumps and features.



**[B16 CE 2204]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**BUILDING PLANNING & DESIGN**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. The Line plan of a residential building is shown in Figure:1. Draw
  - i. Dimensioned Plan
  - ii. Section along AB
  - iii. Front Elevation by following the specifications given below.

Specifications:

1. Foundations: The depth of foundations shall be 1000 mm below ground level. The plain cement concrete (1:4:8) bed in the foundation will be 800 mm wide and 200 mm deep. The footings shall be of brick masonry in C.M (1:4). Width of first and second footing will be 500 mm and 400 mm respectively, whereas depth of both footings will be 400 mm.
2. Basement: The height of basement is 600mm Damp proof course of 50 mm thick shall be provided under the superstructure walls. Thickness of walls in the basement is 300mm
3. Superstructure: The walls in the superstructure will be of brick masonry in C.M (1:6) and all walls except the partition between toilets are 200 mm thick and the partition walls are 100 mm thick from floor. A square brick pillar 200X200 mm is provided at the left corner in front verandah.
4. Lintels and Sunshades: Lintels with R.C.C (1:2:4) are provided on all openings and depth is 150 mm with a bearing of 150 mm on either side. Sun Shades 100 mm thick at the wall face and 75 mm thick at free end are provided. All Sun Shades shall project 600 mm from face of the wall.
5. Height of Super Structure: The Walls in the superstructure are taken to a height of 3300 mm.
6. Roofing: Roofing consists of R.C.C (1:2:4) slab 110 mm thick and weather proof course with two courses of flat tiles in C.M (1:4) 50 mm thick is laid over it.
7. Flooring: Flooring shall be of polished Shahabad Stone slab 25 mm thick over 80 mm thick cement concrete (1:3:6) over sand filling basement.
8. Parapet wall: Parapet wall 100 mm thick and 700 height with brick masonry in C.M (1:4) shall be constructed all around the building.
9. Steps: Steps provided in the front and rear side of length 1200 mm. the width of tread is 300 mm and rise of step is 150mm  
D<sub>1</sub> 1000 X 2100 Panelled Door

|                |                              |
|----------------|------------------------------|
| D <sub>2</sub> | 900 X 2000 Panelled Door     |
| W <sub>1</sub> | 1200 X 1500 Glazed Window    |
| W <sub>2</sub> | 1000 X 1500 Glazed Window    |
| V <sub>1</sub> | 1000 X 600 Glazed Ventilator |

(34 M)

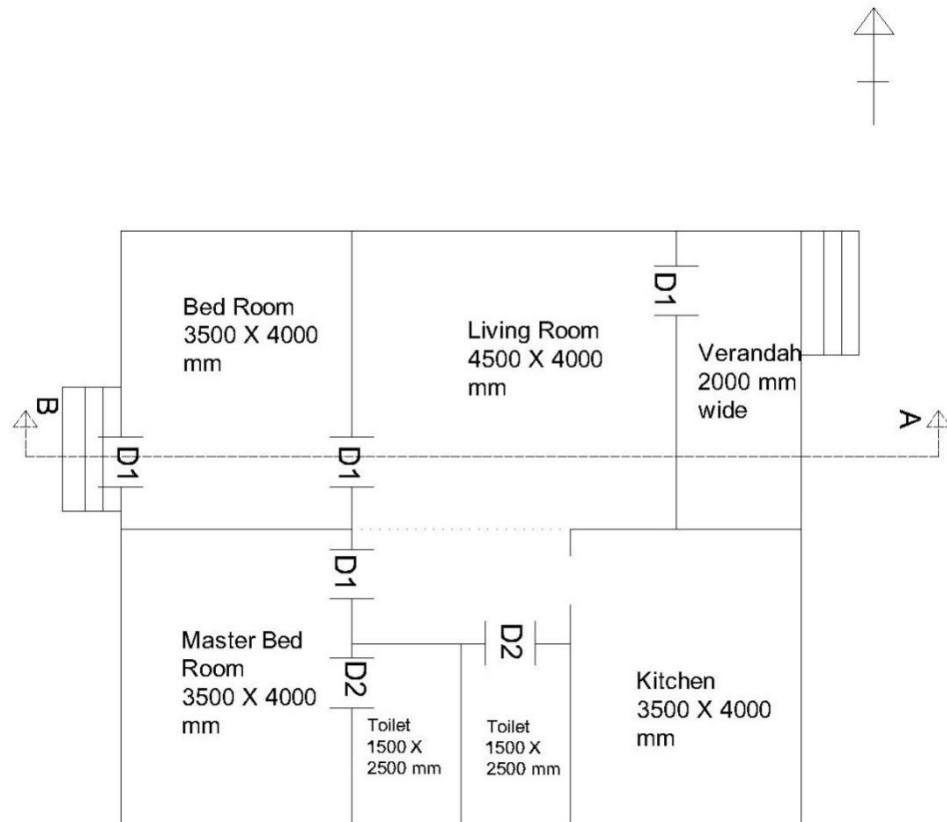


Figure: 1

2. A. What is meant by Orientation of Building? Mention some suggestions for good orientation of the building in a tropical climate (6 M)
- B. What do you understand by “Principles of Planning” of a Building? Explain the significance of Aspect and Prospect for Residential Buildings. (6 M)
3. A. Summarize the various requirements of a Residential Building that affect the Selection of a suitable site for the building. (6 M)
- B. Distinguish between “ Detached “ and “ Semi – Detached Houses “ (6 M)

4. A. What are the usual requirements of a normal Residential Building? (6 M)
- B. Discuss the Various Bye-laws as applied to Buildings and indicate their usefulness (6 M)
5. A. Draw neat sketches to indicate the conventional signs for the following
- i. Brick Masonry
  - ii. Concrete
  - iii. Timber (6 M)
- B. Draw the conventional symbols for
- i. Bed
  - ii. Refrigerator
  - iii. Kitchen Sink (6 M)
6. What is NBC? Write the building bye-laws for open spaces, plinth height and plinth area restrictions for different plot sizes. (12 M)

**[B16 CE 2205]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ADVANCED SURVEYING METHODOLOGIES**  
MODEL QUESTION PAPER  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write short answers for the following:
  - a. What is meant by face left and face right observations? Why is it necessary to take observations on both faces?
  - b. What is Beaman Stadia arc? Mention its use.
  - c. Distinguish between repetition and reiteration.
  - d. Explain the principle of Tacheometry.
  - e. What do you understand by term an ideal curve?
  - f. State the formula to find the length of transition curve.
  - g. What is axis signal correction?
2. Explain the step by step procedure for measuring the horizontal angles by the method of repetition. Mention the advantages and disadvantages.
3. A and B are two stations of a location traverse, their coordinates in m are as follows:

|   | <b>Total Latitude</b> | <b>Total Longitude</b> |
|---|-----------------------|------------------------|
| A | 34,321                | 7,509                  |
| B | 33,670                | 9,652                  |

A Straight reach of railway is to run from B, roughly South of A to B invisible from C and roughly North of B, the offsets perpendicular to the railway being AC = 130m and BD = 72 m. Calculate the bearing of CD.

4. A.) What is Tangential Method of Tacheometry? Explain the different cases in Tangential method of Tacheometry.  
  
B.) Two distances of 20 m and 10 m were accurately measured and the intercepts on the staff between the outer stadia webs were 0.196 m at the former distance and 0.996 m at the later. Calculate the tacheometric constants.
5. A.) Derive the fundamental equation of a Stadia tacheometry.  
B.) In a subtense bar observation, the horizontal angles measured from two stations A and B are  $3^{\circ}10'$  and  $1^{\circ}20'$  respectively. Compute the distance between A and B if the length of subtense bar is 2m.

6. A.) What are the different types of horizontal circular curves? How would you select the most suitable type for a particular case?  
B.) Two Length intersect at a point C of chainage 3506m. The angle of deflection to the right is  $40^{\circ}$ . Two straights are to be connected by a composite curve consisting of a circular curve of radius 320 m and two transition curves of length 50 m each on either side. Calculate the various elements and deflection angles for setting out the transition curve. The peg interval on the circular curve is 20m.
7. A.) Explain the procedure of setting out a curve by Rankines method. Give the computations carried out in this procedure.  
B.) Two tangents intersect at chainage 1180 m with their deflection angle being  $36^{\circ}$ . Adopting the method of deflection angles, calculate the necessary data to set out a simple curve of 30m radius.
8. Write short notes on any THREE:  
v.) Base Line Measurements  
vi.) Elements of Reverse Curve  
vii.) Errors in theodolite.

**[B16 CE 2206]**  
II/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ENGINEERING GEOLOGY**  
**MODEL QUESTION PAPER**  
CIVIL Engineering

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Answer the following:
  - a. Specific gravity of mineral
  - b. Chevron fold and thrust fault
  - c. Draw neat sketch of fault and label its parts
  - d. Draw Schlumberger method electrical Resistivity
  - e. Over break of tunnel
  - f. Geological Time scale
  - g. Strike & dip
2. Describe the landforms produced by running water and describe about its civil engineering importance.
3. What is mineral? How do you identify minerals with help of physical properties of minerals
4. What are igneous rocks? Write about textures, structures and forms of Igneous rocks and write its engineering importance
5. Define the term fold? Describe about classification of folds with neat sketches
6. Explain principles of geophysical methods. How do you identify soil profile using electrical Resistivity method?
7. Write an essay on the role of engineering geologist in planning, design and construction of civil engineering works
8. Describe the geological investigations for dams and reservoirs with neat sketches

**[B16 CE 2206]**

**[B 16 CE 3101]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**STRUCTURAL ANALYSIS**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

1. Write short notes on the following. 14M
  - (a) Describe briefly the general solution procedure underlying the Method of consistent Deformation 2M
  - (b) What are the advantages and disadvantages of Kani's method over moment Distribution Method? 2M
  - (c) Mention any three reasons due to which sway may occur in portal frames. 2M
  - (d) Differentiate kinematic indeterminacy from static indeterminacy? 2M
  - (e) Differentiate between an arch action and beam action. 2M
  - (f) What is function of stiffening girder in a suspension bridge? 2M
  - (g) Distinguish between the flexibility method and stiffness method? 2M
2. Analyze the truss with cross diagonals and two hinged supports shown in Fig. (a) by using Horizontal reaction at A as the redundant. (1000)

Numbers in parentheses are

Areas in  $\text{mm}^2$

$E=200 \text{ kN/mm}^2$

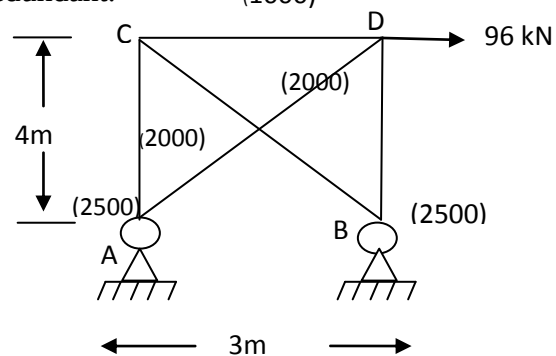


Fig. (a)

3. Analyze the portal frame shown in Fig. (b) by moment distribution method and draw the B.M and S.F diagram.

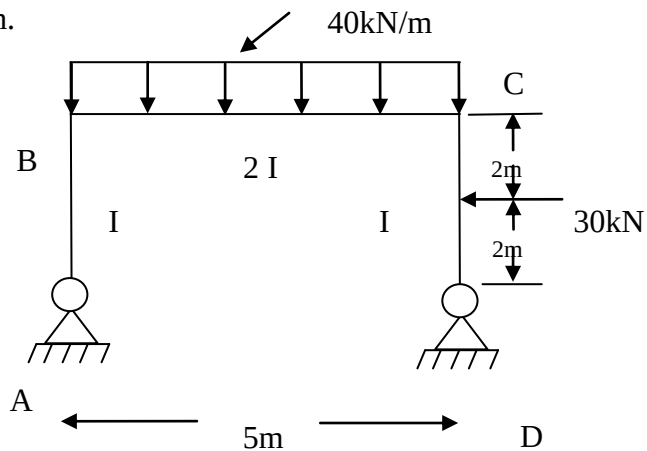


Fig (b)

4. A two hinged parabolic arch has span 40 m and rise of 6 m it has second moment of arch varies as secant of the slope of rib axis and carries uniformly distributed load 30 kN/m over left half of the span together with concentrated load of 120 kN act at 5 m from right support. Calculate the reactions and horizontal thrust at the ends and point out the values of maximum positive and negative moments and also find out the radial shear and normal thrust at 10 m from right support. 14M
5. A suspension cable is suspended from two piers 200 m apart, left support being 5 m above the right. The cable carries a u. d. l of 15 kN/m and has its lowest point 10 m below the lower support. The ends of the cable are attached to saddles on roller at top of piers and the back stays which may be assumed straight are inclined at  $60^\circ$  to the vertical. Determine: 14M
- the maximum tension in the cable
  - the tension in the back stays
  - the thrust on each pier
6. Analyse the portal frame as shown in the Fig. (c), by Kani's method.



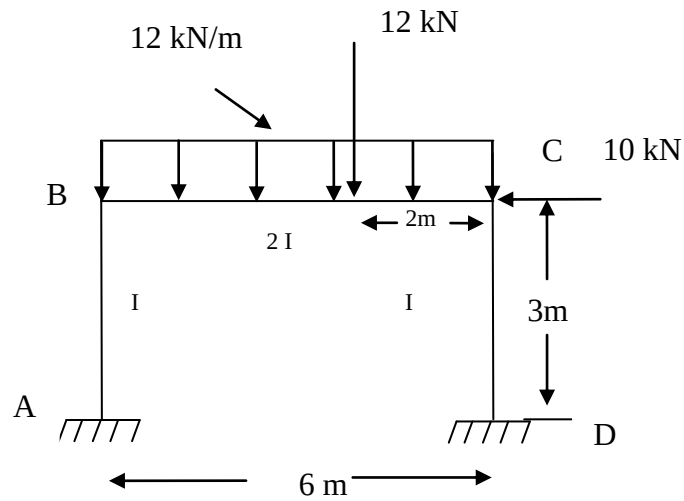


Fig. (c)

7. Analyse the continuous beam shown in the Fig. (d). Using flexibility matrix method.

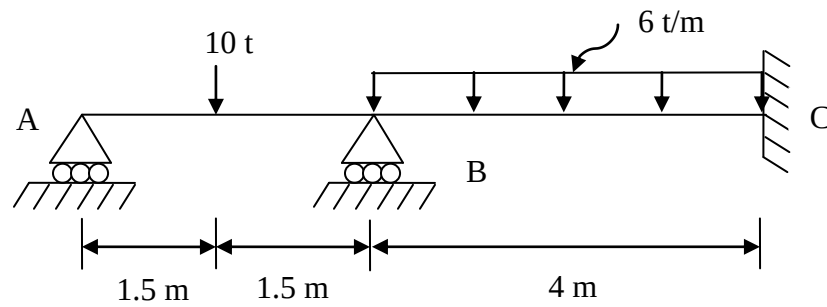


Fig. (d)

8. Analyse the continuous beam shown in the Fig. (e). Using stiffness matrix method.

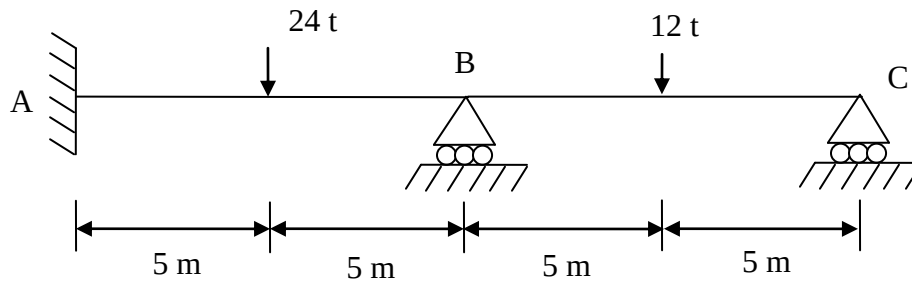


Fig. (e)

**[B16 CE 3102]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**REINFORCED CONCRETE STRUCTURES - II**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

1. a) Distinguish between Cantilever and Counter fort Retaining Walls.  
b) What type of Stresses will be Induced Predominantly if the Water Tank is of Circular Shape.  
c) Classify the different types of bridges.  
d) Distinguish between Bunkers and Silos.  
e) What are the advantages of pile caps.
2. Compute the maximum bending moment and design a cantilever retaining wall to retain earth for a height of 3.6m. The back fill is horizontal. The density of soil is  $18\text{kn/m}^3$ . Safe bearing capacity of soil is  $100\text{kn/m}^2$ . Take the coefficient of friction between concrete and soil as 0.6. The angle of repose is 30 degree. Depth of the foundation is 0.6m. Width of the base slab is 2.4m. Thickness of the stem 200mm. Use  $M_{20}$  grade concrete and Fe-415 steel. Thickness of base slab is 0.3m.
3. Compute the maximum bending moment and design only counter fort and upright slab of counter fort retaining wall retains earth for a height of 7.25m. The density of soil is  $18\text{kn/m}^3$ . Safe bearing capacity of soil is  $200\text{kn/m}^2$ . Take the coefficient of friction between concrete and soil as 0.6. The angle of repose is 30 degree. Depth of the foundation is 1.25m. Width of the base slab is 2.4m. Resisting moment is 2147.5kN-m. Weight is 645.9kN. Thickness of the stem 200mm. Use  $M_{20}$  grade concrete and Fe-415 steel.
4. An intze tank of 9m dia is covered with 100mm thick dome with a central rise of 1.6m. A ring beam is provided at the junction of cylindrical wall and top dome. The height of cylindrical wall is 4.60m which is supported by a conical slab. Maximum water level is up to bottom of ring beam. Compute stresses and design top dome, top ring beam and cylindrical walls.
5. Compute the maximum B.M & S.F in four pile group. Sketch the reinforcement details. Factored load from  $C_d$  of size 600x300mm is 1800kn. pile dia 450mm, pile center to center distance 1.2m.

6. A solid slab bridge has a clear span of 5m and 350mm thick slab. Effective depth is 320mm clear carriage width is 6.80m with 600mm wide kerbs on both sides. Compute the bending moment due to dead load and class 'AA' tracked loading. Assume W.C. is 80mm thick.
7. Compute the maximum bending moment and design the side walls and hopper bottom of a 3m by 3m square bunker to store 30 tonnes of coal. Density of coal is  $9\text{ kN/m}^3$ . Angle of repose is 30 degrees. Adopt  $M_{20}$  grade concrete and Fe-415 HYSD bars. Sketch the details of reinforcement in the bunker.
8. Compute the maximum shear force in a T-beam bridge of span 18m due to class 'AA' tracked loading. The bridge consists of 3 T-beams spaced at 2.60m, are stiffened by 5 cross girders. The clear carriage width is 8.60 m.

**[B16 CE 3102]**

**[B16 CE 3103]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**STEEL STRUCTURES**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer** any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

- |     |                                                                                                                                                                                                                                                                                          |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1.  | Write a short notes on the following.                                                                                                                                                                                                                                                    | 14M |
| (a) | Why are drilled holes preferable than punched holes?                                                                                                                                                                                                                                     | 2M  |
| (b) | What is the main purpose of lacings and battens?                                                                                                                                                                                                                                         | 2M  |
| (c) | Distinguish between bearing bolts and friction grip bolts?                                                                                                                                                                                                                               | 2M  |
| (d) | Draw stress-strain diagram for mild steel mention its salient features?                                                                                                                                                                                                                  | 2M  |
| (e) | What is lug angle? Why lug angles are used?                                                                                                                                                                                                                                              | 2M  |
| (f) | State the advantages of welded connections over riveted connections?                                                                                                                                                                                                                     | 2M  |
| (g) | Differentiate between a slab base and gusset base                                                                                                                                                                                                                                        | 2M  |
| 2.  | Design a tension member to transmit a pull of 150 kN. Effective length is 4.5 m. Member should consist of a pair of angles connected to the one side of the gusset plate of thickness 10 mm. Use 20 mm bolts of grade 4.6                                                                | 14M |
| 3.  | Design a connection to joint two plates of sizes 250 mm × 12 mm of grade Fe 410 to mobilize full plate tensile strength using shop fillet welds if (i) a lap joint is used (ii) a double cover butt joint is used                                                                        | 14M |
| 4.  | Design a column with two channels connected back to back and battened to support a load of 3500 kN. The effective length of the column is 5 m. Design battens also.                                                                                                                      | 14M |
| 5.  | Design a gusseted base for a column ISHB 250 @ 64.96 kg/m two cover plates 300 mm × 20 mm one to each flange to transmit an axial load of 1800 kN. The concrete pedestal is M 20 grade.                                                                                                  | 14M |
| 6.  | Determine the safe load and efficiency of a single cover butt joint. The main plates are 18 mm thick connected by 22 mm diameter bolts at a pitch of 150 mm. Design the cover plate also. What is the percentage reduction in the efficiency of the joint if the plates are lap jointed. | 14M |
| 7.  | Design a laced column 10 m long to carry a factored axial load of 1100 kN.the                                                                                                                                                                                                            | 14M |

column is restrained in position but not in direction at both ends provide single lacing system with bolted connected. Design the column with two channels back to back?

8. Design the base plate for an ISHB 300@ 618 N/m column to carry a factored load of 14M 1000 kN. Assume Fe 410 grade steel and M 25 grade concrete.

**[B16 CE 3103]**

**[B16 CE 3104]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**GEOTECHNICAL ENGINEERING-I**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short note on the following. 14M
  - (a) What are the Index properties of a fine grained soil?
  - (b) List the factors that affect permeability of soils.
  - (c) State Darcy's Law and mention its Limitations.
  - (d) Differentiate between consolidation and compaction.
  - (e) What are the causes of preconsolidation in soils?
  - (f) Define Critical Void Ratio.
  - (g) Write a short note on New mark's Influence Chart.
2. (a) Derive the Relation  $\rho_d = \frac{(1-n_a)G \cdot \rho_w}{1+W_g}$  7M
  - (b) In a field exploration, a soil sample was collected in a sampling tube of internal diameter 5cm below ground water table. The length of the extracted sample was 10.2cm and its weight was 387gm. If  $G=2.7$ , and the weight of the dried sample is 313gm, Determine the porosity, Void Ratio, Degree of Saturation, Bulk Density and dry density of the sample. 7M
3. (a) Discuss about Indian Standard Classification System. 7M
  - (b) A Constant head Permeability test was carried out on a cylindrical sample of sand 10cms dia & 15cm height. 160cm<sup>3</sup> of water was collected in 1.75 mins. Under a head of 30cm. Compute the Coefficient of permeability(K) & Velocity of Flow. 7M
4. (a) Explain the factors Effecting Permeability. 7M
  - (b) A Sand deposit consists of two layers. The top layer is 2.5m thick ( $\rho=1709.67\text{Kg/m}^3$ ) and the bottom layer is 3.5m thick ( $\rho_{\text{sat}}=2064.52\text{Kg/m}^3$ ). The water table is at a depth of 3.5m from the surface and the Zone of capillary saturation is 1 m above the water table, Determine the Total, Neutral and effective 7M

stresses.

5. (a) Explain about Newmark's Influence Chart 7M
- (b) A Concentrated Load of 2000KN is applied at the ground surface. Determine the vertical stress at a point P which is 6m directly below the load. Also calculate the vertical stress at a point R which is at a depth of 6m but at a horizontal distance of 5m from the axis of the load. 7M
6. (a) What is the Effect of Compaction on Engineering Properties of Soil? Discuss in briefly 7M
- (b) Soil Investigation at a site gave the following information. Fine Sand exists to a depth of 10.6m and below this lies a soft clay layer 7.60m thick. The water table is at 4.60m below the ground surface. The submerged unit weight of sand is  $10.4 \text{ KN/m}^3$ , and the wet unit weight above the water table is  $17.6 \text{ KN/m}^3$ . The water content of the normally consolidated clay  $W=40\%$  and Liquid limit  $W_L=45\%$  and  $G=2.78$ . The proposed construction will transmit a net stress of  $120 \text{ KN/m}^2$  at the centre of the clay layer. Find the average settlement of the clay layer. 7M
7. (a) Discuss Terzaghi's theory of consolidation, Stating the Various assumptions. 7M
- (b) A Stratum of Clay is 2m thick and has an initial overburden pressure of  $50 \text{ KN/m}^2$  at its middle. Determine the final settlement due to an increase of  $40 \text{ KN/m}^2$  at the middle of the clay layer. The clay is over-consolidated, with a pre consolidation pressure of  $75 \text{ KN/m}^2$ . The values of the coefficients of recompression and Compression index are 0.05 and 0.25, respectively. Take initial Void ratio as 1.40. 7M
8. (a) Describe the triaxial shear test. What are the advantages of triaxial shear test over the direct shear test. 7M
- (b) A Shear Vane of 7.5cm diameter and 11.00cm length was used to measure the shear strength of a soft clay. If a torque of 600N-m was required to shear the soil, Calculate the shear strength. 7M

The Vane was then rotated rapidly to cause remolding of the soil. The Torque Required in the remoulded state was 200N-m. Determine the Sensitivity of the soil.

**[B16 CE 3105]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**ENVIRONMENTAL ENGINEERING - I**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

1. Write a short note on the following: 14M
  - (a) Distinguish between disinfection and sterilization.
  - (b) What are the objectives of a water supply system?
  - (c) What is temporary hardness and how is it removed?
  - (d) What is blue baby syndrome? What is the pollutant responsible?
  - (e) How is colour imparted to water?
  - (f) Define break point chlorination
  - (g) Differentiate between dry and wet intake
2. (a) What is meant by “per capita demand” of water? Explain how do you estimate water demand for a town or city [7M]  
(b) Using any two methods, calculate the population of a city given below for 2021 and 2031.

|            |          |          |          |          |
|------------|----------|----------|----------|----------|
| Year       | 1981     | 1991     | 2001     | 2011     |
| Population | 1,10,000 | 1,55,000 | 2,06,000 | 2,42,000 |

[7M]
3. (a) What is meant by Infiltration gallery? Explain with a neat sketch, its construction and operation [7M]  
(b) Distinguish between surface water, open well and deep bore wells water with respect to suitability, quantity and quality of water [7M]
4. (a) Explain the harmful effects of excess turbidity, chlorides and nitrates in water. [7M]  
(b) Mention different types of pipes and pipe materials and write the merits and demerits. [7M]
5. (a) What are the various chlorination practices and write the significance of break point chlorination. [7M]  
(b) Explain EDTA method of analysis by titrimetry. [7M]
6. (a) Explain the purpose of pumping in a distribution system. [7M]  
(b) With a neat sketch, explain the working of a rapid sand filter. [7M]



7. (a) Design a slow sand filter for the given data:  
Population = 60000  
Rate of water supply = 135 lpcd  
Rate of filtration = 300 l/m<sup>2</sup>/day  
L/B ratio = 2:1. [7M]
- (b) Write short notes on various water borne diseases. [7M]
8. (a) Explain the methods of removal of odours from water. [7M].  
Why is their removal necessary?
- (b) Explain about the “Nalgonda Technique” for removal of fluorides [7M].

**[B16 CE 3105]**

**[B16 CE 3106]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**CONCRETE TECHNOLOGY**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

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**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**  
**All parts of a question must be answered at one place only**

- |     |                                                                                                  |     |
|-----|--------------------------------------------------------------------------------------------------|-----|
| 1.  | Write short notes on the following.                                                              | 14M |
| (a) | Define Initial setting time and Final setting time                                               | 2M  |
| (b) | Deleterious substance in aggregate                                                               | 2M  |
| (c) | What is AAR? Write its significance.                                                             | 2M  |
| (d) | Define workability. Mention the names of different types of workability tests                    | 2M  |
| (e) | Draw a neat sketch for testing of flexural concrete specimen.                                    | 2M  |
| (f) | What is meant by 'Creep in concrete'?                                                            | 2M  |
| (g) | Acceptance criteria                                                                              | 2M  |
| 2.  | (a) What are Bogue's compounds? Explain their significance                                       | 5 M |
|     | (b) List out the laboratory tests for cement. Explain determination of setting times of cement.  | 10M |
| 3.  | (a) Explain AAR. What are the factors promoting AAR? Explain.                                    | 7M  |
|     | (b) Explain bulking of fine aggregate. How do you determine it?                                  | 7M  |
| 4.  | (a) Explain workability of concrete and its influencing factors. Explain any one test in detail. | 7M  |
|     | (b) Explain how the setting time of concrete will be determined.                                 | 7M  |
| 5.  | (a) Describe briefly the process of manufacture of Concrete                                      | 7M  |
|     | (b) Explain in brief Segregation and Bleeding                                                    | 7M  |
| 6.  | (a) List out different Non-Destructive evaluation techniques to assess the strength of Concrete. | 7M  |
|     | (b) Explain Radioactive method of assessment of strength of concrete.                            | 7M  |

7. (a) Explain the elastic properties of concrete. 7M  
(b) Explain the shrinkage of concrete. How it is classified? Explain any one of them. 7M
8. (a) Design mix proportions with the following data using IS code method. Characteristic compressive strength of concrete 25Mpa. 20mm Maximum size crushed aggregate. Medium degree of workability. Specific gravity of cement 3.10. Specific gravity of coarse and fine aggregates 2.72 and 2.61. Zone II sand. Average quality control. 14M

**[B16 CE 3106]**

**[B16 CE 3107]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**GIS and REMOTE SENSING**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**  
**Answer any FOUR questions from the remaining.**  
**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

- |    |      |                                                                                |    |
|----|------|--------------------------------------------------------------------------------|----|
| 1. | (a). | Define Remote Sensing.                                                         | 2M |
|    | (b). | Explain electromagnetic radiation interaction with atmosphere.                 | 2M |
|    | (c). | What are the advantages and disadvantages of various Remote Sensing platforms. | 2M |
|    | (d). | Write a note on classification of Sensors.                                     | 2M |
|    | (e). | Evaluate the elements of visual interpretation in image analysis.              | 2M |
|    | (f). | Compare supervised and unsupervised classification.                            | 2M |
|    | (g). | Evaluate the digital image processing techniques.                              | 2M |
| 2. | (a). | Expand GIS. Describe the key components of GIS in detail with a neat sketch.   | 7M |
|    | (b). | Distinguish the raster data models and vector data models.                     | 7M |
| 3. | (a). | Explain the role of Map Projections in GIS.                                    | 7M |
|    | (b). | Briefly explain the application areas of GIS.                                  | 7M |
| 4. | (a). | Demonstrate the role of Remote Sensing in forest studies.                      | 7M |
|    | (b). | Explain the significance of Remote Sensing and GIS in Urban Planning.          | 7M |
| 5. | (a). | Explain the function of Remote Sensing in Geological studies.                  | 7M |
|    | (b). | Demonstrate the role of Remote Sensing in Land Use Land Cover studies.         | 7M |
| 6. | (a). | Explain the role of Remote Sensing for Watershed Management.                   | 7M |
|    | (b). | Illustrate the role of RS & GIS in Flood zoning and its mapping.               | 7M |
| 7. | (a). | Elucidate the role of GIS for Environmental Impact Assessment.                 | 7M |
|    | (b). | GIS for Ground Water prospects mapping - Explain.                              | 7M |
| 8. | (a). | Explain in detail about the concept of resolution                              | 7M |
|    | (b). | Discuss in brief about spatial and radiometric resolutions.                    | 7M |

**[B16 CE 3107]**

**[B16 CE 3108]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**PRESTRESSED CONCRETE**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

1. a) For what types of structures do you recommend post-tensioning? [14M]  
b) What are the factors influencing the creep and shrinkage of concrete?  
c) What is the necessity of using high strength concrete in pre stressed concrete?  
d) Explain about anchorage slip.  
e) What is 'Effective reinforcement Ratio'?  
f) Explain Long term deflection of pre stressed concrete beams.  
g) What is Guyon's method of approach
  
2. a) Briefly explain the relation between tendon profiles and equivalent loads in prestressed concrete beams with sketches. [7M]  
b) A Prestressed concrete beam of rectangular cross section 300mm by 600mm is 10 m long supports a live load 15 kN/m in addition to its own self weight. The beam is prestressed by a cable having high-tensile wires of 2000 mm<sup>2</sup> area stressed to 1000 N/mm<sup>2</sup>. The cable is straight and located at a distance of 175 mm from the soffit of the beam. Determine the shift in the pressure line at one quarter span and centre of span, when the beam supports the service load. [7M]
3. A pretensioned beam, 200mm wide and 300mm deep, is prestressed by 10 wires of 7mm diameter initially stressed to 1200N/mm<sup>2</sup>, with their centroids located 100mm from the soffit. Find the maximum stress in concrete immediately after transfer, allowing only for elastic shortening of concrete. Take  $E_s = 210 \text{ kN/mm}^2$ ,  $E_c = 5700(f_{cu})^{1/2}$ , Creep=1.6, shrinkage strain =  $3 \times 10^{-4}$  [14M]
4. A prestressed beam of 8m span has a symmetrical I-section. The flange are 200mm width and 60mm thick. The web thickness is 80mm and overall depth of girder is 400mm. The member is prestressed by 8 wires of 5mm diameter located on the tension side such that the eccentricity is 90mm. The internal stress in the wires is 1280N/mm<sup>2</sup> and the cube strength at transfer is 42N/mm<sup>2</sup>. Determine the maximum vertical tensile stress developed in the transfer zone, and design suitable mild steel reinforcement, assuming the permissible stress in steel as 140N/mm<sup>2</sup>. [14M]
  
5. a) Explain the various modes of failure encountered in prestressed concrete beams subjected to bending moment, shear and torsion. [7M]

b) A concrete beam of rectangular section, 300 mm wide and 800 mm deep is subjected to a twisting moment of 40 kN m and a prestressing force of 150 kN acting at an eccentricity of 220 mm. Calculate the maximum principal tensile stress. If the beam is subjected to a bending moment of 120 kN m in addition to the twisting moment, calculate the maximum principle tensile stress. 6. A solid slab bridge has a clear span of 5m and 350mm thick slab. Effective depth is 320mm clear carriage width is 6.80m with 600mm wide kerbs on both sides. Compute the bending moment due to dead load and class 'AA' tracked loading. Assume W.C. is 80mm thick. [7M]

6. a) Explain with sketches 'Hoyer's long line system of pretensioning' [7M]

b) A prestressed concrete girder is post-tensioned using a cable concentric at supports and having an eccentricity of 400 mm at centre of span. The effective span of the girder is 20m. The initial force in the cable is 400 kN at the jacking end A. Determine the loss of force in the cable due to friction and wave effect and the effective force in the cable at the farther end B. Assume coefficient of friction = 0.30 and coefficient for wave effect  $K = 0.0043/\text{m}$ . [7M]

7. A prestressed concrete pile, 250 mm square, contains 60 pre-tensioned wires, each of 2mm diameter, uniformly distributed over the section. The wires are initially tensioned on the prestressing bed with a total force of 300 kN. Calculate the final stress in concrete and the percentage loss of stress in steel after all losses, give the following data:

$E_s = 210 \text{ kN/mm}^2$

$E_c = 32 \text{ kN/mm}^2$

Shortening due to creep =  $30 \times 10^{-6} \text{ mm/mm per N/mm}^2$

Total shrinkage =  $200 \times 10^{-6} \text{ per unit length}$

Relaxation of steel stress = 5 per-cent of initial stress

Prestressing force,  $P = 300 \text{ kN}$

[14M]

8. a) What are the advantages of using composite construction with prestressed and in situ concrete in structural members? [7M]

b) Briefly outline the method of computing the ultimate flexural and shear strength of composite sections. [7M]

**[B16 CE 3108]**

**[B16 CE 3109]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**BUILDING SERVICES AND MAINTENANCE**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

- |    |                                                                                                  |     |
|----|--------------------------------------------------------------------------------------------------|-----|
| 1. | Write short notes on the following.                                                              | 14M |
|    | (a) Differentiate expansion and contraction joints?                                              | 2M  |
|    | (b) Define grouting and write about its importance in structures?                                | 2M  |
|    | (c) What is active crack?                                                                        | 2M  |
|    | (d) Write about distempering?                                                                    | 2M  |
|    | (e) What is meant by 'Shear key'?                                                                | 2M  |
|    | (f) What is meant by 'Creep in concrete'?                                                        | 2M  |
|    | (g) What are the advantages of providing D. P. C to the existing buildings.                      | 2M  |
| 2. | (a) Differentiate white washing and colour washing                                               | 7 M |
|    | (b) List out the various problems occurring in building service life.                            | 7 M |
| 3. | (a) Explain repair of broad cast reception installation system.                                  | 7M  |
|    | (b) Explain electrical installation system?                                                      | 7M  |
| 4. | (a) Explain about the drain and sanitary works                                                   | 7M  |
|    | (b) Write importance of re-polishing of terrazzo and mosaic.                                     | 7M  |
| 5. | (a) Describe briefly the repair of joints                                                        | 7M  |
|    | (b) Explain the removal of stains from concrete and terrazzo floor                               | 7M  |
| 6. | (a) List out various types of cracks in concrete and explain in detail.                          | 7M  |
|    | (b) Explain possible damages to the structural element-beams and slab.                           | 7M  |
| 7. | (a) State and explain the repair of fire services system.                                        | 7M  |
|    | (b) Explain the strengthening of foundation and foundation soil.                                 | 7M  |
| 8. | (a) Explain the stepwise procedure for repair of columns using jacketing method with neat sketch | 14M |

**[B16 CE 3109]**

**[B16 CE 3110]**  
III/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**WATERSHED MANAGEMENT**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only

- |       |                                                                                                                                     |       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 a)  | What is the function of a watershed?                                                                                                | [2M]  |
| b)    | Bring out the role of peoples participation in watershed management.                                                                | [2M]  |
| c)    | List out the different types of erosion.                                                                                            | [2M]  |
| d)    | What are the consequences of soil erosion?                                                                                          | [2M]  |
| e)    | What are the various types of rain water harvesting?                                                                                | [2M]  |
| f)    | Write a short note on land grading operation.                                                                                       | [2M]  |
| g)    | What is the role of ecosystem?                                                                                                      | [2M]  |
|       |                                                                                                                                     |       |
| 2) a) | Explain in detail the importance of watershed management in India.                                                                  | [7M]  |
| b)    | Define watershed development programme. Also Explain how integrated and multi disciplinary approach will help watershed management. | [7M]  |
|       |                                                                                                                                     |       |
| 3) a) | Explain Gumbell method of flood frequency analysis.                                                                                 | [7M]  |
| b)    | Explain various methods of runoff calculation.                                                                                      | [7M]  |
|       |                                                                                                                                     |       |
| 4) a) | What is meant by term erosion? Explain the causes of erosion.                                                                       | [7M]  |
| b)    | Discuss in detail about contour techniques and rock fill dams erosion control methods.                                              | [7M]  |
|       |                                                                                                                                     |       |
| 5) a) | What are the basic steps to begin the process of rain water harvesting ?                                                            | [7M]  |
| b)    | Describe the role of check dam, farm ponds and percolation tanks in rain water harvesting                                           | [7M]  |
|       |                                                                                                                                     |       |
| 6. a) | Explain land use and land capability. List out the different objectives of land capability classification.                          | [7M]  |
| b)    | Explain in detail about the classification of land capability?                                                                      | [7M]  |
|       |                                                                                                                                     |       |
| 7. a) | Explain in detail inter, mixed and strip cropping.                                                                                  | [7M]  |
| b)    | Give a brief explanation on social forestry and afforestation.                                                                      | [7M]  |
|       |                                                                                                                                     |       |
| 8.    | Write about importance and application of RS and GIS in watershed management                                                        | [14M] |

**[B16 CE 3110]**



**[B16 CE 3201]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ADVANCED STEEL STRUCTURES**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

- |     |                                                                                                                                                                                                                                                              |     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1.  | Write a short notes on the following.                                                                                                                                                                                                                        | 14M |
| (a) | What is the difference between a beam and a plate girder?                                                                                                                                                                                                    | 2M  |
| (b) | When do you go for plate girder web splicing?                                                                                                                                                                                                                | 2M  |
| (c) | What is an end post.                                                                                                                                                                                                                                         | 2M  |
| (d) | What are the functions of transverse intermediate stiffeners and bearing stiffeners?                                                                                                                                                                         | 2M  |
| (e) | Distinguish between laterally supported and unsupported beams?                                                                                                                                                                                               | 2M  |
| (f) | How does the plate girder derive post-buckling strength                                                                                                                                                                                                      | 2M  |
| (g) | Distinguish between unstiffened and stiffened seat angle connections.                                                                                                                                                                                        | 2M  |
| 2.  | Design a laterally unsupported beam with 6 m simply supported effective span, subjected to a point load of 40 kN at mid span, select a section of ISMB 300 mm.                                                                                               | 14M |
| 3.  | Design a welded plate girder of 20 m span using the tension field action for the following factored forces. Maximum Moment $M_z = 5000$ kN, Maximum shear Force $V_z = 900$ kN. The girder is laterally restrained, connections need not be designed         | 14M |
| 4.  | Design an elevated cylindrical steel tank with hemispherical bottom, for a capacity of 1,20,000 liters. The tank has conical roof and its ring beam is 15 m high above the G.L. Take basic wind pressure as $1.5 \text{ kN/m}^2$ ?                           | 14M |
| 5.  | Design a laterally supported beam with 6 m simply supported effective span, subjected to a point load of 40 kN at mid span. Select a section of ISMB 300 mm.                                                                                                 | 14M |
| 6.  | Design an unstiffened welded seat connections for beam <u>ISMB 250@365.9</u> N/m transmitting an end reaction of 110 kN, due to the factored loads the flange of the column ISHB 200 @ 365.9 N/m. The seat angle is welded to the column flange in workshop. | 14M |
| 7.  | Design the plate girder using intermediate transverse stiffeners. Connections need not be designed. Use post critical method for the design. Maximum bending moment 11,448 kNm and shear force 1908 kN.                                                      | 14M |

8. Design a welded plate girder 24 m in span and laterally restrained throughout. It has to support a uniform load of 100 kN/m throughout exclusive of self weight. Design the girder without intermediate transverse stiffeners. Design the cross section with end load bearing stiffeners and connections. 14M

**[B 16 CE 3201]**

**[B16 CE 3202]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**GEOTECHNICAL ENGINEERING-II**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write a short note on the following. 14M
  - (a) What are the factors effecting the bearing capacity of foundations?
  - (b) Write short notes on Depth of Exploration.
  - (c) Write a short notes on classification of piles based on mode of Transfer of load.
  - (d) What is meant by grip length? What is the importance in well foundation
  - (e) Discuss about different factors of safety in Stability of Slopes.
  - (f) Write a short notes on types of retaining walls.
  - (g) What are the measures to reduce the settlement
2. (a) Explain wash boring with neat sketch. 7M  
(b) A Square footing is to be designed to carry a total load of 500KN in a soil having  $C=10\text{KN/m}^2$ ,  $\phi=25^\circ$ ,  $\gamma=16\text{KN/m}^3$ . The footing is to be laid at a depth of 2mts. below ground level and F.S is 3. Find the Dimension of the footing. Given  $N_c=22$ ,  $N_q=11$ ,  $N_\gamma=9$ . 7M  
**(CO 2, K 3)**
3. (a) Derive the expression for Terzaghi's Bearing Capacity theory and write the assumptions? 7M  
(b) Determine the safe bearing capacity of a square footing  $1.2\text{m} \times 1.2\text{m}$  at a depth of 1.5 m in a soil with  $\gamma_{sat} = 19.5 \text{ kN/m}^3$ , if the water table can rise up to ground surface. Take  $C = 18 \text{ kN/m}^2$  and  $\phi = 30^\circ$ . Assume general shear failure conditions. Use Terzaghi's theory. For  $\phi = 30^\circ$ ,  $N_c = 37.2$ ,  $N_q = 22.5$  and  $N_\gamma = 19.7$ . 7M
4. (a) Explain pile load test with neat sketch. 7M  
(b) A concrete pile, 25cm in diameter is driven in to a dense sand ( $\phi = 30^\circ$ ,  $\gamma=20\text{kN/m}^3$ ,  $K=1$ ,  $\tan \delta = 0.70$ ) for a depth of 6m. If the water table is at 2m from the ground level, estimate the safe load, taking Factor of safety 3. 7M
5. (a) Explain Pneumatic Caisson with neat sketch and mention its advantages and disadvantages. 7M

- |        |                                                                                                                                                                                                                                                                                                                       |    |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| (b)    | Discuss the Causes and Remedies for Tilts and Shifts (with neat sketches) in Well Foundations.                                                                                                                                                                                                                        | 7M |
| 6. (a) | Discuss the Friction circle method for the stability analysis of slopes. Can this method be used for purely cohesive soil?                                                                                                                                                                                            | 7M |
| (b)    | A cut of length 10m is made in a cohesive soil deposit ( $C=30\text{kN/m}^2$ , $\phi = 0^\circ$ , $\gamma=20\text{kN/m}^3$ ). There is a hard stratum under the cohesive soil at a depth of 12m below the original ground surface. If the required factor of safety is 1.5, determine the safe slope                  | 7M |
| 7. (a) | What are the assumptions of Rankine's theory? Derive the expressions for active pressure and passive pressure.                                                                                                                                                                                                        | 7M |
| (b)    | Determine the passive pressure by Rankine's theory per unit run for a retaining wall 4m high, with $i=15^\circ$ , $\phi = 30^\circ$ and $\gamma=20\text{kN/m}^3$ . The back face of the wall is smooth and vertical.                                                                                                  | 7M |
| 8. (a) | Explain types of slope failures.                                                                                                                                                                                                                                                                                      | 7M |
| (b)    | A pile group consists of 9 piles of 300mm dia and length 10m, is arranged in 3 rows at a spacing of 750mm c/c. The piles are driven into a deposit of clay whose properties are $C=100\text{kN/m}^2$ , $\gamma=20\text{kN/m}^3$ . Determine the safe load on the pile group. Take $F.S=3$ , $\alpha =0.6$ , $N_c=9$ . | 7M |

**[B16 CE 3202]**

**[B16 CE 3203]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**FLUID MECHANICS - III**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. (a) Explain the term dimensionally homogeneous equation.  
(b) Define Froude number and Mach number.  
(c) What is Slip of a reciprocating pump? Why does it occur?  
(d) What is Net Positive Suction Head?  
(e) What is a Draft tube? What are its functions?  
(f) Define Air Vessel with a neat sketch.  
(g) What is the function of Guide Vanes in the Francis turbine?

2. a) The Resistance  $R$ , to the motion of a supersonic aircraft of Length  $L$ , moving with a Velocity  $V$ , in air of density  $\rho$ , depends on the Viscosity  $\mu$  and bulk modulus of elasticity  $K$  of air. Obtain using Buckingham  $\pi$ -theorem, the following expression for the resistance  $R$ .

$$R = \rho L^2 V^2 f\left(\left(\frac{\mu}{\rho V L}\right), \left(\frac{K}{\rho V^2}\right)\right) \quad (7M)$$

b) A 1:10 scale model of a submarine moving far below the surface of water is tested in a water tunnel. If the speed of the prototype is 8m/sec. Determine the corresponding velocity of water in the tunnel. Also determine the ratio of drag for model and prototype.

$$\begin{array}{ll} \mu_{\text{sea water}} = 1.121 \times 10^{-6} \text{ m}^2/\text{sec} & \mu_{\text{water}} = 1.00 \times 10^{-6} \text{ m}^2/\text{sec} \\ \rho_{\text{sea water}} = 1027 \text{ kg/m}^3 & \rho_{\text{water}} = 1000 \text{ Kg/m}^3 \end{array} \quad (7M)$$

3.(a) Define Specific Speed of a Hydraulic Turbine and derive an expression for the Turbine. (7M)

(b) A Pelton Wheel has a mean bucket speed of 12m/sec and is supplied with water at rate of 750lit/sec under a head of 35m. If the bucket deflects the jet through an angle of  $160^\circ$ , find the power developed by the turbine and its Hydraulic Efficiency. Take  $C_v = 0.98$  neglect friction losses. Also determine Overall Efficiency ( $\eta_o$ ) and its mechanical efficiency ( $\eta_{\text{mech}}$ ) is 80%. (7M)

4. (a) The diameters of an impeller of a Centrifugal pump at inlet and outlet are 30cm and 60cm respectively. Determine the Minimum Starting Speed of the pump if it works against a head of 30m. (7M)

- (b) explain any three Similarity Laws with practical applications. (7M)
5. (a) Give a comparative analysis of Centrifugal pump and Reciprocating pump. (7M)
- (b) A double-acting reciprocating pump, running at 50 r.p.m. is discharging 900 litres of water per minute. The pump has a stroke of 400mm. The diameter of piston is 250mm. The delivery and suction heads are 25m and 4m respectively. Find the Slip of the pump and Power required to drive the pump. (7M)
6. (a) Define Indicator Diagram. How will you prove that area of Indicator Diagram is proportional to the work done by Reciprocating Pump? (7M)
- (b) A Centrifugal Pump discharges 7500 litres of water per minute against a total head of 25m when running at 660rpm. The outer diameter of the impeller is 60 cm and ratio of outer to inner diameter is 2. The area of flow through the wheel is  $600\text{cm}^2$ , the vanes are set back at an angle of  $45^\circ$ , water enters the wheel radially. Calculate Manometric Efficiency ( $\eta_{\text{mano}}$ ) and Vane angle at inlet  $\theta$ . (7M)
7. (a) Derive an expression for force exerted by a jet on stationary curved plate if jet strikes the curved plate at the centre. (7M)
- (b) Classify the Characteristic Curves of Centrifugal Pump with neat sketches. (7M)
8. (a) Define Hydraulic Ram and explain its working. (7M)
- (b) Design a Francis turbine runner with following data:
- Net Head  $H=68\text{m}$ , Speed  $N=750\text{ rpm}$ , Output Power  $P=330\text{ KW}$ ,  $\eta_o=85\%$ ,  $\eta_h=94\%$ , flow ratio  $\psi=0.15$ , breadth ratio  $n=0.1$ , inner diameter of the runner is  $(1/2)$  the outer diameter, velocity of flow remains constant throughout and is radial at exit. (7M)

**[B16 CE 3203]**

**[B16 CE 3204]**  
**III/IV B.Tech. DEGREE EXAMINATION**  
**Second Semester**  
**ESTIMATING AND QUALITY SURVEYING**  
**MODEL QUESTION PAPER**  
**DEPARTMENT OF CIVIL ENGINEERING**

**Time: 3 Hrs.**

**Max. Marks: 70**

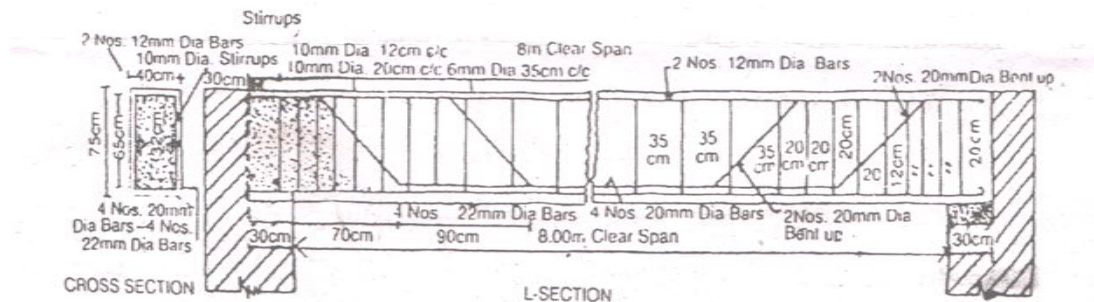
**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Answer all the following.
  - a) State the methods of preparing approximate estimates.
  - b) Distinguish between scrap value and salvage value?
  - c) Define Gross income, Net income.
  - d) List out the different methods of estimating the earth work?
  - e) What do you understand by standard schedule of rates?
  - f) What are the units of measurement for
    - (i) Pointing (ii) DPC (iii) Rolling shutters (iv) Stone masonry
  - g) What is the main purpose of rate analysis?
2. Prepare the rate analysis for the following items of work. Assume suitable rates for material and labour.
  - a) Brick masonry work in CM (1:6)
  - b) Plastering in CM (1:4), 12mm thick for walls
3. Prepare a detailed estimate of the following items for a R.C.C. beam of 8 meter clear span and 75cm X 40cm in section from the given drawing

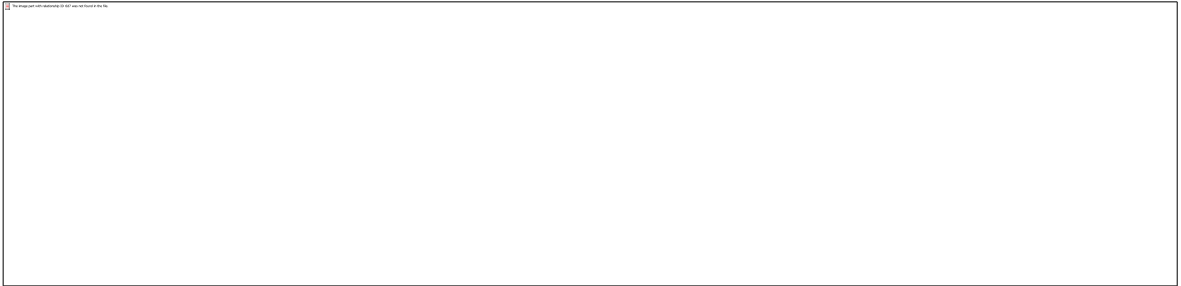


*RCC Rectangular beam*

- (a) R.C.C. work including centering and shuttering
- (b) Steel in detail shall be calculated separately and

(c) Prepare a schedule of bars.

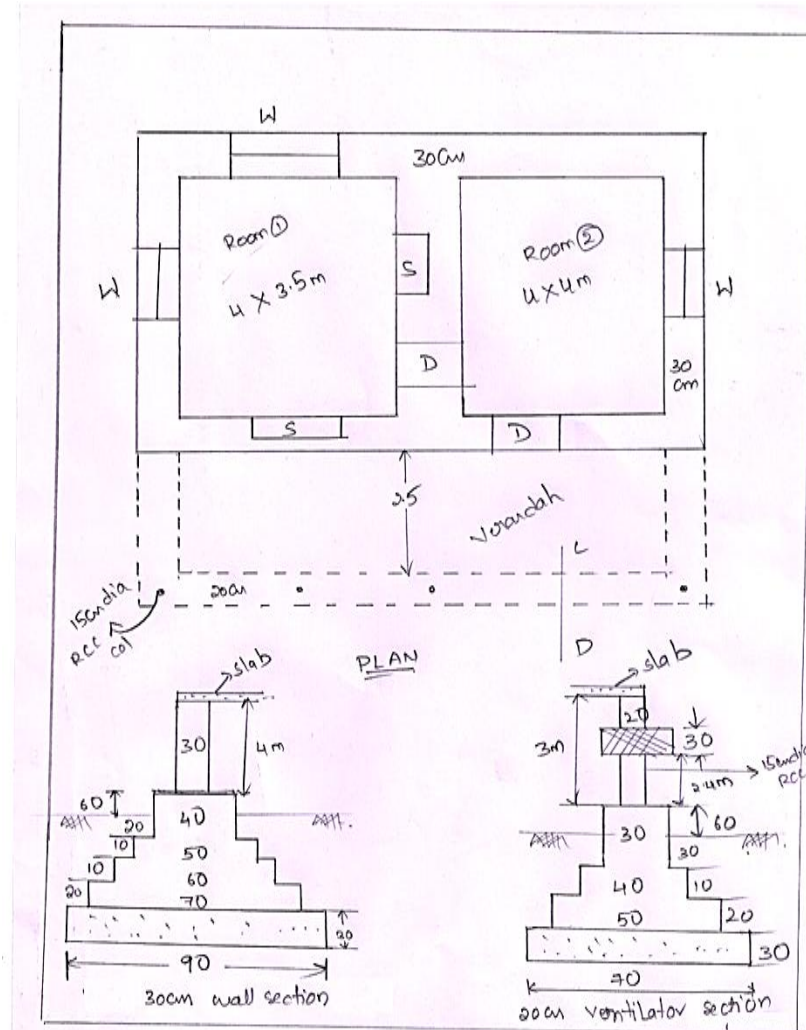
4. Estimate the cost of earthwork for a portion of road for 360m length from the following data:



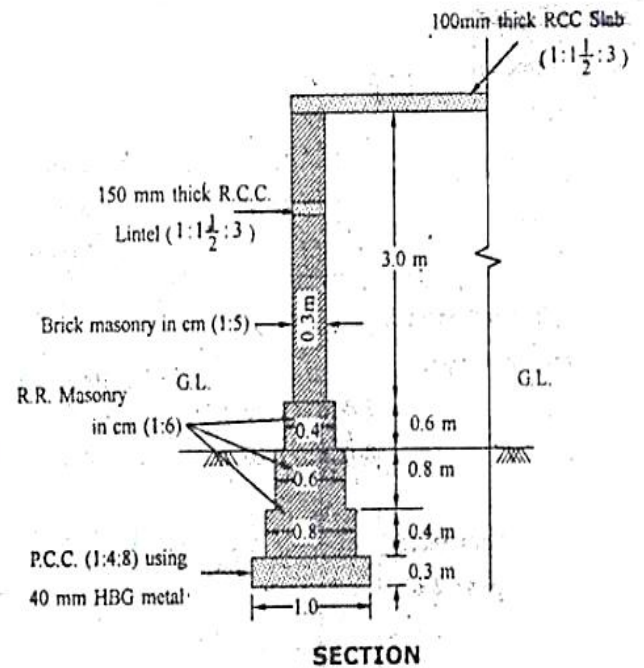
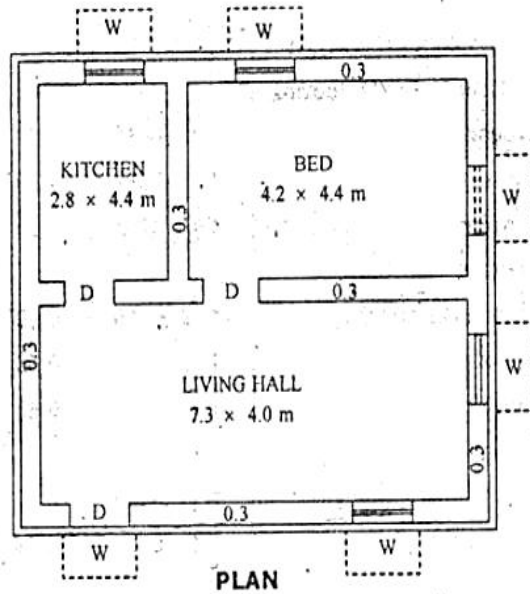
The formation width of the road is 10m and the side slopes are 2:1 in banking and 1.5:1 in cutting

5. Estimate the quantities of the building items of a two roomed building from the given plan and section as shown in Figure:
- a) Lime concrete in foundation
  - b) 1st class brick work with 1:6 cement mortar in foundation and plinth





6. Prepare the detailed estimate for the following items of works from the given Figure by using Long wall and Short wall method:
- P.C.C. (1:4:8) using 40 mm HBG metal for foundations.
  - R.C.C. (1:1.5:3) for lintels and R.C.C slab.



7. A freehold property having an area of 800 sq.m. is jointly held by four brothers and it is fully developed. It's a 4 storeyed building having a basement. The structure is being designed and used as a college building. It is an RCC framed structure given for a monthly rent of Rs. 8000/- (for whole building). The usual outgoings may be taken as 20% of gross annual rent. Work out the share of each brother in property. (Assume missing data wherever necessary)
8. Give the detailed specifications for the following items of work
  - a) RCC (1:2:4) for roof slabs
  - b) Wood work for door and window frames

**[B16 CE 3205]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ENVIRONMENTAL ENGINEERING - II**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write a short note on the following: [14M]
  - (a) Distinguish between CBOD and NBOD.
  - (b) Differentiate sewage from sullage.
  - (c) State the operational troubles of trickling filters.
  - (d) What is the difference between manhole and drop manhole?
  - (e) What is RBC?
  - (f) What is the principle of imhoff tank design?
  - (g) What are the factors affecting the quantity of sewage?
2. (a) Discuss different sewerage systems along with their relative merits and demerits [7M]
  - (b) Explain house plumbing systems with neat sketches. [7M]
3. (a) Explain functions of gully trap and floor trap with neat sketches. [7M]
  - (b) Explain the working of pumping station in detail with a neat sketch [7M]
4. (a) Explain the factors affecting the design of a storm water sewer. [7M]
  - (b) Explain the working of Grit Chamber with a neat sketch. [7M]
5. (a) Differentiate aerobic and anaerobic decomposition in detail. [7M]
  - (b) Explain the working of trickling filter with a neat sketch. [7M]
6. (a) Determine the size of a high rate trickling filter for the following data:  
Flow = 4.0 Mld  
Recirculation ratio = 1.5  
BOD of raw sewage = 220 mg/L  
BOD removed in primary clarifier = 30%  
Final effluent desired = 50 mg/L  
Calculate also the size of the standard rate trickling filter to accomplish the above requirement. [7M]
  - (b) What is meant by activated sludge process? Explain the properties of activated sludge and its action. [7M]
1. (a) What is sedimentation tank? Explain its working principle and operation in detail. [7M]

- (b) What is a stabilization pond? Discuss the working principle and enumerate the various types of stabilization ponds. [7M]
2. (a) List the objectives of sludge digestion. Explain various methods of sludge treatment processes. [7M]
- (b) Design a septic tank for 200 users. [7M]

**[B16 CE 3205]**

**[B16 CE 3206 ]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**GROUND IMPROVEMENT TECHNIQUES**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

- |    |                                                                                                            |     |
|----|------------------------------------------------------------------------------------------------------------|-----|
| 1. | Write a short note on the following.                                                                       | 14M |
|    | (a) What are various methods of in situ densification of cohesive soils?                                   |     |
|    | (b) Define grouting. List any two applications of grouting.                                                |     |
|    | (c) What are the four major applications of geotextiles?                                                   |     |
|    | (d) What are the components of reinforced earth?                                                           |     |
|    | (e) Write a short note on soil bitumen stabilization.                                                      |     |
|    | (f) What do you understand by preloading?                                                                  |     |
|    | (g) What is a stone column? and explain its function                                                       |     |
| 2. | (a) Explain the impact at ground surface method of densifying granular soils.                              | 7M  |
|    | (b) Explain the electro osmotic method of dewatering for ground improvement.                               | 7M  |
| 3. | (a) Explain in detail the vacuum well point system of dewatering.                                          | 7M  |
|    | (b) Describe the vibroflotation technique of densifying granular soil.                                     | 7M  |
| 4. | (a) Explain in detail with the help of a neat sketch the different stages of grouting                      | 7M  |
|    | (b) Define grouting. What are the objectives of grouting?                                                  | 7M  |
| 5. | (a) What are the design principles of reinforced earth wall?                                               | 7M  |
|    | (b) What are the factors governing the design of reinforced earth walls?                                   | 7M  |
| 6. | (a) Explain in detail the use of geosynthetics as a reinforcement                                          | 7M  |
|    | (b) Explain different functions of geotextiles with neat sketches.                                         | 7M  |
| 7. | (a) Explain in detail the mechanical soil stabilization.                                                   | 7M  |
|    | (b) Explain the principle and application of soil-lime stabilization                                       | 7M  |
| 8. | (a) Explain how the engineering properties of soil are changed by the process of bituminous stabilization. | 7M  |
|    | (b) What is a stone column? What are the methods of installing a stone column?                             | 7M  |

**[B16 CE 3207]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**ENVIRONMENTAL IMPACT ASSESSMENT**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

1. Write short notes on the following: [14 M]
  - (a) What are environmental attributes?
  - (b) What is Leopold matrix?
  - (c) What is cost-benefit analysis?
  - (d) Define environmental setting.
  - (e) Differentiate between network and matrix methods
  - (f) What do you understand by the term "Screening in the EIA process"?
  - (g) What are the impacts of water pollution?
2. (a) Differentiate between EIA and EIS. Explain how do you prepare EIS? [7M]  
(b) Explain the Ad Hoc and checklist method. [7M]
3. (a) Explain in detail step by step procedure for conducting EIA. [7M]  
(b) Discuss the significance of environmental setting in an EIA study. [7M]
4. (a) How do you identify various potential environmental impacts resulting from an engineering project? [7M]  
(b) Explain the Battelle environmental evaluation system for water projects. [7M]
5. (a) Write the EIA of a thermal power plant in brief. [7M]  
(b) Explain the criteria for selection of EIA methodology. [7M]
6. (a) What are different attributes of EIA? Explain any one of them. [7M]  
(b) Write the EIA report of a hydraulic power plant. [7M]
7. (a) Outline the basic steps for prediction and assessment of impacts on the air environment. [7M]  
(b) What is EMP? Explain its components. [7M]
8. (a) Explain the environmental impacts of National Highway on the environment. [7M]  
(b) Explain the impact of pharmaceutical industry on environment. [7M]

**[B16 CE 3207]**

**[B16 CE 3208]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**MARINE STRUCTURES**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

- |    |                                                                                      |     |
|----|--------------------------------------------------------------------------------------|-----|
| 1. | Write a short note on the following.                                                 | 14M |
|    | (a) What is meant by storm surge?                                                    |     |
|    | (b) What is the purpose of offshore structures?                                      |     |
|    | (c) What are the breakwater effects on beach?                                        |     |
|    | (d) What are the environmental loads on offshore structures?                         |     |
|    | (e) What are the advantages of semi submersibles                                     |     |
|    | (f) What do you understand by soil structure interaction?                            |     |
|    | (g) What are the uses of gravity platforms                                           |     |
| 2. | (a) Explain the Design principles of Rubble mound and composite Breakwaters.         | 7M  |
|    | (b) Differentiate the wharves and jetties.                                           | 7M  |
| 3. | (a) Explain in detail the Gravity Platforms Guyed Towers.                            | 7M  |
|    | (b) Explain in detail the Articulated Towers.                                        | 7M  |
| 4. | (a) What are the considerations involved in selection of site and type of breakwater | 7M  |
|    | (b) Define offshore structures. Classify various types of offshore structures?       | 7M  |
| 5. | (a) What are the wave forces on small and large cylinders? Explain in detail?        | 7M  |
|    | (b) What are the factors governing the design of stability of rubble structures?     | 7M  |
| 6. | (a) Explain in detail the Tension Leg Platforms                                      | 7M  |
|    | (b) Explain briefly about Waves, tides, tsunamis, storm surge and currents.          | 7M  |
| 7. | (a) Explain in detail the procedure for the installation of jacket platforms.        | 7M  |
|    | (b) Explain the preliminary design aspects of offshore structures.                   | 7M  |
| 8. | (a) Explain briefly about the components of sea wall.                                | 7M  |
|    | (b) What are the design considerations of breakwaters and jetties?                   | 7M  |

**[B 16 CE 3209]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**URBAN HYDROLOGY**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

**Question No. 1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

**All parts of a question must be answered at one place only**

- |    |                                                                                                                                   |     |
|----|-----------------------------------------------------------------------------------------------------------------------------------|-----|
| 1. | Write a short notes on the following.                                                                                             | 14M |
|    | (a) What is meant by IDF curves?                                                                                                  | 3M  |
|    | (b) What is meant by Urban hydrology cycle?                                                                                       | 3M  |
|    | (c) List out the elements of drainage systems?                                                                                    | 3M  |
|    | (d) What is comprehensive planning in drainage system?                                                                            | 3M  |
|    | (e) What is the use of models in planning drainage?                                                                               | 2M  |
| 2. | What are the methods of estimation of time of concentration for design of urban drainage systems                                  | 14M |
| 3. | What are the elements of drainage system and explain the necessity of open channel?                                               | 14M |
| 4. | What is the interrelation between water resources investigation and urban planning processes and write about planning objectives. | 14M |
| 5. | What are the storm water drainage structures, and design of storm water network                                                   | 14M |
| 6. | What is meant by Urban drainage system and Design storms for urban drainage systems                                               | 14M |
| 7. | What are constructed wetlands? And discuss about models available for storm water management?                                     | 14M |
| 8. | Write a short notes on                                                                                                            | 14M |
|    | i) Underground drains                                                                                                             |     |
|    | ii) Quality and quantity of runoffs                                                                                               |     |

**[B16 CE 3209]**



**[B16 CE 3210]**  
III/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**FINITE ELEMENT METHODS**  
MODEL QUESTION PAPER  
DEPARTMENT OF CIVIL ENGINEERING

**Time: 3 Hrs.**

**Max. Marks: 70**

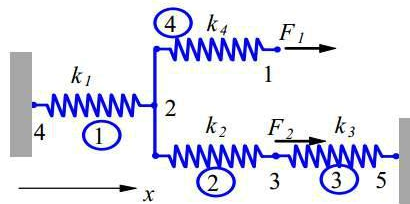
**Question No.1 compulsory.**

**Answer any FOUR questions from the remaining.**

**All Questions Carry equal marks**

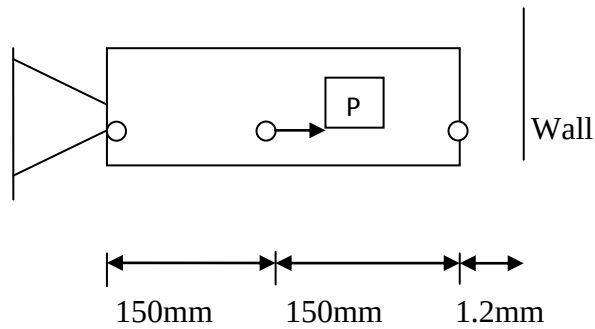
**All parts of a question must be answered at one place only**

1. Write short notes on the following. 14M
  - (a) Explain about plane stress 2M
  - (b) Define shape function 2M
  - (c) Difference between CST and LST element 2M
  - (d) What are the advantages of FEM 2M
  - (e) Element stiffness matrix for bar and beam element 2M
  - (f) Define interpolation function 2M
  - (g) What is a global stiffness matrix 2M
2. (a) For the spring system shown in the figure find the displacements at the nodes and the reactions. Given  $K_1=200$  N/mm,  $K_2=K_4=150$  N/mm,  $K_3= 200$  N/mm,  $F_1=500$  N,  $F_2=600$ N. 10M

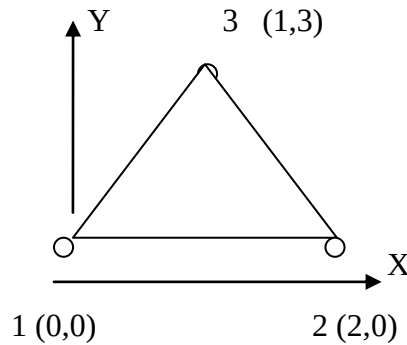


- (b) Explain about discretization of element 4M
3. (a) Derive shape function for 1D bar element. 7M
  - (b) Derive the stiffness matrix of a truss element. 7M
4. A rod subjected to an axial load  $P=600$ Kn is applied as shown in fig. determine the following 14M
  - a) Displacements at each node.

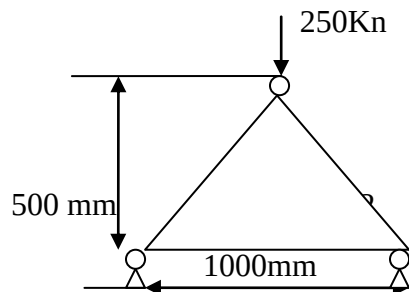
b) stresses at each element.



5. Calculate the stiffness matrix the temperature force vector elements. The element experiences a  $20^{\circ}\text{C}$  increase in temperature. Assume coefficient of thermal expansion is  $6 \times 10^{-6}/^{\circ}\text{C}$ . The co-ordinates are given in MM's. Assume plane stress conditions. Take  $E = 2.1 \times 10^5 \text{ N/mm}^2$ , Poisson's ratio = 0.25,  $t = 10\text{mm}$ . 14M



6. (a) Derive the Strain-Displacement Matrix [B] for Axisymmetric triangular element 10M  
 (b) Derive plain strain condition in CST Elements 4M
7. Calculate the nodal displacement, stress and reactions at the supports, for given truss member. Take  $E = 2 \times 10^5 \text{ N/mm}^2$  and area of members  $A_1 = A_2 = A_3 = 2500\text{mm}^2$  14M



8. A fixed beam of length AC 8m carries a UDL of  $w$  (N/m) which run over a length 14M  
AB of 4m from fixed end. Calculate the rotation at point B.

**[B16 CE 3210]**

**[B16CE4101]**  
IV/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**WATER RESOURCES ENGINEERING-I**  
**DEPARTMENT OF CIVIL ENGINEERING**  
MODEL QUESTION PAPER

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only.

1. Write short notes on the following:
  - a. Explain the terms 'duty' and 'delta' 2M
  - b. Compare Kennedy's and Lacey's silt theories 2M
  - c. Distinguish between Darcy velocity and actual velocity 2M
  - d. Define hydrograph 2M
  - e. Recommend two measures to reduce reservoir evaporation 2M
  - f. What are the advantages of crop rotation? 2M
  - g. What is balancing depth? 2M
2. Discuss with a neat sketch various method used to compute average rainfall over a basin 14M
3.
  - a. The rates of rainfall for the successive 30 min period of a 3-hour storm are: 1.6, 3.6, 5.2, 2.2, 1.0 cm/hr. The corresponding surface runoff is estimated to be 3.6cm. Establish the  $\phi$ -index. Also determine the W-index 7M
  - b. Explain briefly the infiltration process. What are the factors that influence the process of infiltration? 7M
4.
  - a. An aquifer has an average thickness of 50m and an aerial extent of 150 ha, Estimate the available ground water storage if the aquifer is unconfined and fluctuation in GWT is observed as 10m. 7M
  - b. State Darcy's law and its limitations 7M
5.
  - a. Explain clearly the I.S.D. curves method of reservoir flood routing. 7M
  - b. What are the factors to be considered in choosing the routing period? 7M
6.
  - a. The root zone of a certain soil has a field capacity of 25% and permanent wilting percentage is 8%.  
What is the depth of moisture in the root zone at field capacity and permanent wilting point? 7M
  - b. How much water is available if the root zone depth is 1.1m? The dry weight of the soil is  $13.75 \text{ kN/m}^3$  7M
7.
  - a. Describe Lacey's theory for the design of irrigation channel in alluvial soil 7M
  - b. Design an irrigation channel to carry a discharge of 6 cumec. Assume  $N=0.0225$  and  $m=1$ .  
The channel has a bed slope of 0.25 m per kilometer 7M
8. Describe the step by step procedure of the derivation of a unit hydrograph from an isolated storm 14M

**[B16CE4101]**

**[B16CE4102]**  
IV/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**TRANSPORTATION ENGINEERING - I**  
**DEPARTMENT OF CIVIL ENGINEERING**  
MODEL QUESTION PAPER

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only.

1. Write short notes on the following:
  - a. Briefly explain Factors controlling Highway alignment? 2M
  - b. What are the important factors on which the highway geometric design depends? Explain briefly. 2M
  - c. Explain spot speed, running speed, Space-Mean speed and Time Mean Speed? 2M
  - d. List the tests on bitumen emulsion and mention the objects of each. 2M
  - e. Discuss about Equivalent single wheel load. 2M
  - f. Mention the causes of distress in flexible pavements and explain any two with neat sketches. 2M
  - g. Briefly outline the points of difference on 'Flexible and Rigid pavements' 2M
2.
  - a. Discuss about the second twenty year road plan (1961-81) and its salient features? 7M
  - b. (i) Briefly outline the main features of various road patterns commonly in use? 7M  
(ii) From the following data for a district, calculate the road length required based on Nagpur road plan.

$\text{Total area} = 6300 \text{ km}^2$   
 $\text{Agricultural area} = 2800 \text{ km}^2$

No. of villages with population ranges <500, 501-1000, 1001-2000, 2001-5000, and above 5001 are 450, 320, 110, 50 and 10 respectively. Length of railway track = 75 km.
3.
  - a. (i) Discuss briefly about Kerbs and their designs. 7M  
(ii) Calculate the safe overtaking sight distance for a design speed of 96 kmph. Assume all other data suitably.
  - b. Derive an expression for finding the extra widening required on horizontal curve. 7M
4.
  - a. Explain how the speed and delay studies are carried out. What are the various uses of speed and delay studies? 7M
  - b. (i) How are O & D study data represented and interpreted? 7M  
(ii) Spot speed studies were carried out at a certain stretch of a highway with mixed traffic flow and the consolidated data collected are given below.

| Speed range, kmph | No. of vehicles observed | Speed range, kmph | No. of vehicles observed |
|-------------------|--------------------------|-------------------|--------------------------|
| 0 to 10           | 16                       | 50 to 60          | 270                      |
| 10 to 20          | 22                       | 60 to 70          | 130                      |
| 20 to 30          | 76                       | 70 to 80          | 56                       |
| 30 to 40          | 98                       | 80 to 90          | 42                       |
| 40 to 50          | 240                      | 90 to 100         | 12                       |

Determine: (i) the upper and lower speed limits and (ii) the design speed for checking the geometric design elements of the highway.

5.
  - a. What are the factors on which the strength characteristics of soil depend? 7M
  - b. List different tests on road aggregates and mention their advantages and limitations. 7M
  
6.
  - a. Enumerate various methods of flexible pavement design. Briefly indicate the basis of design in each case? 7M
  - b. Using the data given below, calculate the wheel load stresses at (i) interior, (ii) edge and (c) corner regions of a cement concrete pavement using Westergaard's stress equations. Also determine the probable location where the crack is likely to develop due to corner loading. 7M  
 Wheel load,  $P = 5100 \text{ kg}$ ; Modulus of elasticity of cement concrete,  $E = 3.0 \times 10^5 \text{ kg/cm}^2$ ; Pavement thickness,  $h = 18 \text{ cm}$ ; Poisson's ratio of concrete,  $\mu = 0.15$ ; Modulus of subgrade reaction,  $K = 6.0 \text{ kg/cm}^3$ ; Radius of contact area,  $a = 15 \text{ cm}$ .
  
7.
  - a. Mention the steps for the construction of new highway (i) on embankment (ii) on cutting. 7M
  - b. Explain the features, characteristics and advantages of the following types of overlays over CC pavements: (i) bituminous pavement layers (ii)un – bonded CC overlay (iii) partially bonded CC overlay. 7M
  
8.
  - a. Enumerate the various factors while selecting a suitable site for an Airport? 7M
  - b. The length of a runway at sea-level, standard atmospheric conditions and zero gradient is 1500m. The airport site has an elevation of 900m, and the reference temperature as  $20^\circ \text{ C}$ . If the proposed runway grading permits an effective gradient of 0.20 percent, determine the actual runway length required at the site. 7M

**[B16 CE 4102]**

**[B16 CE 4103]**  
IV/IV B.Tech. DEGREE EXAMINATION  
First Semester  
**PROJECT PLANNING AND MANAGEMENT**  
**CIVIL ENGINEERING**  
MODEL QUESTION PAPER

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only.

1. Write short notes on the following:
  - a. Discuss the important aspects of project planning? 2M
  - b. What is security deposit? 2M
  - c. List out various time estimates of PERT? 2M
  - d. State various conditions of a Contract 2M
  - e. Define the term “Project Management”? 2M
  - f. What are the disadvantages of a Bar chart? 2M
  - g. Explain the term “Resource levelling”? 2M
2.
  - a. Explain “Mile stone Charts” with an Example. 7M
  - b. Draw the Bar chart for the following data: 7M

| Activities     | A | B | C | D | E | F |
|----------------|---|---|---|---|---|---|
| Duration(Days) | 2 | 1 | 3 | 3 | 1 | 2 |

- (i) Activity B starts only after the completion of activity A
  - (ii) Activity B and C can be started simultaneously
  - (iii) Activity D starts 1 day after Activity B starts
  - (iv) Activity E starts after completion of activity D
  - (v) Activity E and F ends at the same timeFind the project duration.
3. Differentiate between the following
  - a. Activities on Arrows (AOA) and Activities on Nodes (AON) network diagrams. 7M
  - b. CPM and PERT techniques. 7M
4.
  - a. Explain the significance of Project Management? 7M
  - b. Discuss various qualities of a manager? 7M
5. A Project consists of nine activities and the details about them are given below in the table.

|                   |   |   |   |   |   |   |   |    |         |
|-------------------|---|---|---|---|---|---|---|----|---------|
| Activity          | A | B | C | D | E | F | G | H  | I(Last) |
| Duration in weeks | 8 | 6 | 3 | 7 | 5 | 6 | 3 | 10 | 5       |
| Predecessor(s)    | - | A | B | C | D | B | B | G  | E,F,H   |

- a. Draw the project network, number the events 7M
  - b. Identify critical path and determine-duration of the project. 7M
6. a. Determine the optimum duration and associated cost of a project consisting of eight activities including one dummy activity. Indirect cost Rs. 80/- per day. The normal and crash time in days and cost in rupees for different activities are given in table shown below. Also draw the least-cost time scaled network. 7M

| Activity | Normal Time (Days) | Crash Time (Days) | Normal Cost (Rs.) | Crash Cost (Rs.) |
|----------|--------------------|-------------------|-------------------|------------------|
| 1-2      | 2                  | 2                 | 1,000             | 1,000            |
| 1-3      | 7                  | 3                 | 500               | 900              |
| 2-3      | 6                  | 3                 | 300               | 420              |
| 2-4      | 5                  | 4                 | 200               | 250              |
| 3-4      | 0                  | 0                 | 0                 | 0                |
| 3-5      | 9                  | 4                 | 600               | 900              |
| 4-6      | 11                 | 6                 | 600               | 1000             |
| 5-6      | 6                  | 3                 | 700               | 910              |

- b. Explain Resource smoothening and Resource Levelling? 7M
7. a. What is a Contract? State different types of Contract and explain any two of them? 7M
- b. What is Arbitration, explain the role of Arbitrator in Dispute Resolution? 7M
8. a. State various deposits made by a Contractor before starting a project? Explain? 7M
- b. Define “Tender” and “Tender Document”? Explain how a project is allocated to a bidder? 7M

**[B16 CE 4103]**



**[B16 CE 4201]**  
IV/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**WATER RESOURCES ENGINEERING II**  
**DEPARTMENT OF CIVIL ENGINEERING**  
MODEL QUESTION PAPER

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only.

1.
  - a. What is surge tank? State its foundation. 2M
  - b. Compare between weir and barrage. 2M
  - c. Under what circumstances, earthen dams are preferred. 2M
  - d. What are the different types of cross drainage works? 2M
  - e. What is a elementary profile of a gravity dam? 2M
  - f. What is a siphon spill way? 2M
  - g. What are groynes? State their functions 2M
2.
  - a. Discuss in detail causes of failure of earth dams. Explain the control measures generally adopted for controlling such failures. 7M
  - b. Explain with neat sketches the different forces that may act on a gravity dam. Indicate their magnitudes, directions and locations. 7M
3.
  - a. Briefly outline Khosla's theory on the design of weirs on permeable foundation. Enumerate the various corrections that are needed in the application of this theory. 7M
  - b. Briefly explain how the Ogee Spillway profile is plotted. 7M
4.
  - a. What are stilling basins? Highlighting their functions, explain with aid of a neat sketch, the location of chute blocks and end sill in a stilling basin 7M
  - b. Explain the method of fixing the waterway of drain in a aqueduct 7M
5.
  - a. Explain the layout of a power house including the super structure 7M
  - b. What is a fall in a canal? How are the canal falls classified? 7M
6.
  - a. What do you understand by a head regulators? State the functions of a distributing head regulator and cross regulator. 7M
  - b. Distinguish between run-off river power plants and storage power plants. Explain the functions of the Intake structure. 7M

- |    |    |                                                                                           |     |
|----|----|-------------------------------------------------------------------------------------------|-----|
| 7. | a. | Discuss briefly the causes of failure of weirs on permeable foundation and their remedies | 7M  |
|    | b. | Explain the following river training works.                                               | 7M  |
|    |    | <b>i)</b> Cutoffs                                                                         |     |
|    |    | <b>ii)</b> Lunching aprons                                                                |     |
|    |    | <b>iii)</b> Guide banks                                                                   |     |
| 8. |    | Write short notes on                                                                      | 14M |
|    |    | <b>a)</b> USBR and IS standard basins                                                     |     |
|    |    | <b>b)</b> Foundation treatment of gravity dam                                             |     |
|    |    | <b>c)</b> Water hammer pressure.                                                          |     |

**[B16 CE 4201]**

**[B16 CE 4202]**  
IV/IV B.Tech. DEGREE EXAMINATION  
Second Semester  
**TRANSPORTATION ENGINEERING - II**  
**DEPARTMENT OF CIVIL ENGINEERING**  
MODEL QUESTION PAPER

**Time: 3 Hrs.**

**Max. Marks: 70**

Question No. 1 compulsory.

Answer any FOUR questions from the remaining.

All Questions Carry equal marks

All parts of a question must be answered at one place only.

1. Write short notes on the following:
  - a. Classify Indian Railways. Discuss the advantages of railways? 2M
  - b. Differentiate between ordinary fish plate and combination fish plate. 2M
  - c. Differentiate between (i) staggered joint and suspended joint. (ii) suspended joint and supported joint 2M
  - d. What is cant deficiency? State two reasons why cant deficiency is limited. 2M
  - e. What do you mean by coning of wheels? 2M
  - f. What is the basic difference between natural harbor and artificial harbor? 2M
  - g. What is meant by Mucking? How it is different from Hauling Muck? 2M
2. a. State the principle requirements of an ideal permanent way. Draw the typical cross section of a railway track for a double line B.G. 7M
3. a. What are the possible causes of creep? What are the effects of creep? Explain the various preventive and remedial measures that can be taken. 7M  
b. If the wheel base of a vehicle moving on a B.G track is 6m, the diameter of wheel is 1.5m and the depth of flanges below the top of rail is 3.17cm. Determine the extra width required to be provided on gauge, if the radius of the curve is 160m. 7M
4. a. Explain the necessity of gradients. Discuss all types of gradients giving their permissible values adopted to Indian Railways. 7M  
b. What are the various rail failures? Discuss them with neat sketches. 7M
5. a. What are the functions of interlocking? Discuss various methods of interlocking. 7M  
b. What is meant by crossing? What are the essential requirements of a good crossing? Discuss various types of crossings in use on Indian Railways. 7M
6. a. State the types of dredging devices commonly used in harbor engineering practice and explain with neat sketches? 7M  
b. What are the navigational aids? Explain them briefly 7M

7. a. Explain with a neat sketch, the procedure of ventilation by combination of blowing and exhausting. 7M
- b. Name the various methods of tunneling in hard and soft rocks. Describe one in each case. 7M
8. a. State the natural phenomena a harbour engineer has to study and briefly mention the effects of these phenomena? 7M
- b. Explain the different types of break waters. 7M

**[B16 CE 4202]**